

Application of Built-in Adaptive Remeshing and Mesh Refinement Features in Abaqus to Fracture Simulations Using Phase-field User Elements

Comprehensive Supervisor Progress Report – 02 September 2026

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Master Thesis in Computational Materials Science
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02 September 2026

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Abstract

This work investigates the application of Abaqus built-in error-guided mesh refinement and adaptive remeshing capabilities to phase-field fracture simulations implemented with user elements. The point of departure is the staggered phase-field implementation of Molnár and Gravouil and the mesh-refinement procedure proposed by Pandey and Kumar. The core research objectives comprise: (1) reproducing reference fracture simulations without mesh refinement; (2) implementing the Abaqus-native refinement procedure in Python while retaining the layered UEL/UMAT/facsimile representation required by the user-element formulation; (3) quantitatively reproducing the reference results with adaptive refinement; (4) integrating and verifying the IMFD ABAQUSER post-processing tool; (5) applying the adaptive workflow to fracture benchmarks and sensitivity studies; and (6) assessing nonmatching state transfer and restart mechanics across evolving discretizations.

First, a fixed-mesh Mode-I single-edge-crack benchmark was reproduced without mesh refinement (Job 1398090.**mmaster02**, 15,192 Quad4 elements, walltime 06 h 31 min, CPUT 06 h 30 min). The reconstructed response achieved a peak reaction force of $F_{\text{peak}} = 0.757778 \text{ kN}$ at $u_{\text{peak}} = 0.005857 \text{ mm}$ compared to the published target of 0.7580 kN at 0.005860 mm, exhibiting relative differences of only -0.029% in peak load and -0.051% in peak displacement. This fixed-mesh calculation established the quantitative baseline for the refinement study (Task 3: **COMPLETE / SCIENTIFICALLY ACCEPTED**).

Second, an Abaqus-native Python-controlled two-job pre-refinement workflow was implemented following Pandey and Kumar. A coarse pre-analysis supplies the Abaqus recovered-stress discretization error indicator MISESeri; an Abaqus **RemeshingRule** and **adaptiveRemesh** operation generate the refined physical mesh; and the resulting connectivity is converted automatically into coincident phase-field UEL, mechanical UEL, and visualization UMAT layers. Qualification testing demonstrated genuine native remeshing and eliminated an implementation defect where calling **Part.generateMesh()** after **adaptiveRemesh()** had previously overwritten the refined mesh with coarse part seeds (Task 4: **COMPLETE / VERIFIED END-TO-END**).

Third, production adaptive Mode-I simulations were performed and reconciled. Reconstructing the nominal 1.0% error target reported in the literature generated an over-refined mesh of 71,320 physical elements (69,443 Quad4, 1,877 Tri3) on a sharp crack seam, which completed technically (Job 1399632.**mmaster02**, walltime 35 h 08 min, CPUT 34 h 03 min) but failed quantitative reproduction gates due to an elastic compliance shift and accelerated crack-tip damage localization ($F_{\text{peak}} = 0.478203 \text{ kN}$, -36.91%). A controlled forensic audit established that because the relative stress error $\eta = \text{MISESeri}/\text{MISESAVG}$ is scale-invariant across a singular crack tip, the nominal 1.0% threshold marks $> 80\%$ of the domain for refinement in our native implementation. While this mechanism explains our observed over-refinement, the exact cause of why the publication reported 13,941 elements cannot be uniquely determined from published information and is classified as **DOCUMENTED_UNRESOLVABLE_FROM_PUBLISHED_INFORMATION**. An empirical 2.0% error-target reconstruction yielded 15,396 physical elements (+10.4% match to the reported $\approx 13,941$ elements). Authoritative production solve 1400395.**mmaster02** completed 7,028 increments with zero cutbacks and zero warnings (**Exit status** 0, walltime 07 h 06 min,

CPUT 06 h 52 min), yielding $K_0 = 137.8437 \text{ kN/mm}$ (-0.08%), $F_{\text{peak}} = 0.748197 \text{ kN}$ (-1.29%), $u_{\text{peak}} = 0.005775 \text{ mm}$ (-1.45%), and a 98.31% post-peak load drop, passing all five pre-declared Task-5 scientific acceptance gates (Task 5: **QUANTITATIVE REPRODUCTION PASSED**). An accompanying controlled 5.0% sensitivity solve (1400396.mmmaster02, 4,194 elements, walltime 01 h 55 min, CPUT 01 h 50 min) yielded $F_{\text{peak}} = 0.764964 \text{ kN}$ ($+0.92\%$) and $u_{\text{peak}} = 0.007060 \text{ mm}$ ($+20.48\%$), demonstrating that coarser crack-corridor refinement diffuses the phase damage gradient and delays peak localization.

Fourth, visualization integration (Task 6) was investigated. In the absence of accessible authentic IMFD ABAQUSER software, an in-solver companion facsimile UMAT visualization bridge was executed under production Job 1400408.mmmaster02 (walltime 07 h 23 min, CPUT 07 h 08 min). The bridge maps UEL phase and history variables through Fortran COMMON /CB_STATE_TRANS/ and UEXTERNALDB lifecycle logic into passive CPE4/CPE3 UMAT elements, writing STATEV(15) = d and STATEV(16) = $\mathcal{H}$ directly to the standard Abaqus output database. The calculation demonstrated exact 0.000000% mechanical parity with baseline 1400395.mmmaster02 across all 7,028 increments. Evaluated phase-field outputs revealed a bounded numerical overshoot ($\max(d) = 1.00518084$), which was confirmed to originate in the solved UEL phase field rather than the visualization transfer (FORMULATION_LEVEL_NOT_VISUALIZATION_TRANSFER, classified as POSSIBLE_FORMULATION_DISCRETIZATION_CAUSE). The companion bridge is **FULLY VERIFIED**, while formal integration of authentic ABAQUSER remains designated as TASK6_BLOCKED_EXTERNAL_ABAQUSER_DEPENDENCY pending software provision by the institute; Task 6 is therefore not scientifically complete.

Finally, nonmatching state transfer and multi-resolution restart mechanics were investigated, demonstrating runtime state ingestion while identifying residual mechanical-continuity challenges upon phase release across changing mesh topologies. The current shared-state UEL architecture is qualified strictly for serial execution due to mutable common storage. With Task 5 accepted and Task 6 verified at the companion-bridge level, the investigation advances to Task 7 benchmark and sensitivity studies.

Chapter 1

Problem Statement, Theoretical Basis, and Reference Methods

1.1 Problem statement and thesis scope

Phase-field fracture approaches regularize sharp discontinuities over a finite diffuse length scale ℓ_0 , replacing discrete interface tracking with the solution of a continuous differential equation for a scalar damage or phase-field parameter $d \in [0, 1]$. While this variational regularization provides a mathematically elegant framework capable of predicting complex fracture phenomena such as crack initiation, branching, merging, and curving without ad-hoc geometric tracking criteria, it introduces a severe spatial resolution constraint: the finite element discretization must resolve the diffuse crack corridor with element sizes typically satisfying $h \leq \ell_0/2$ or $h \leq \ell_0/5$.

When applied to large structural domains, uniform fine meshing becomes computationally prohibitive because fine elements are placed throughout regions that remain purely elastic. This Master’s thesis investigates the systematic application of Abaqus built-in adaptive remeshing and mesh refinement capabilities to phase-field fracture simulations implemented via user-defined finite elements (UEL/UMAT).

The research is governed by the following authoritative proposal work packages:

- Task 1: PFF familiarization:** Theoretical and algorithmic verification of the phase-field fracture implementation.
- Task 2: Literature review:** Systematic analysis of variational phase-field formulations, Abaqus user subroutines, adaptive mesh refinement, and visualization methods.
- Task 3: Reproduce reference simulations without refinement:** Reproduce published benchmark fracture results using available user code on fixed reference meshes.
- Task 4: Implement native mesh refinement in Python:** Implement in Python the mesh refinement / adaptive remeshing methodology described in Pandey & Kumar (Pandey and Kumar, 2025), using Abaqus internal refinement capabilities and required interaction with the PFF user implementation.
- Task 5: Reproduce reference results with mesh refinement:** Reproduce and quantitatively validate the reference-paper results with adaptive mesh refinement enabled.
- Task 6: Integrate ABAQUSER visualization:** Integrate and verify the IMFD ABAQUSER visualization tool (Roth et al., 2014) in the modeling workflow.

Task 7: Apply to fracture benchmarks and sensitivity studies: Apply the verified adaptive remeshing workflow to standard fracture benchmarks and controlled sensitivity analyses.

Task 8: Formulate meshing-parameter recommendations: Derive robust meshing guidelines and error-target selection criteria for future engineering users.

Task 9: Document workflow for future users: Prepare comprehensive technical documentation and user guides.

Task 10: Write thesis: Finalize the Master’s thesis manuscript.

1.2 Governing phase-field fracture formulation

The implementation follows the standard second-order AT2 variational regularization of brittle fracture (Msekh et al., 2015; Molnár and Gravouil, 2017; Pandey and Kumar, 2025; Diddige et al., 2025). Consider an elastic body occupying domain $\Omega \subset \mathbb{R}^2$ with boundary $\partial\Omega$. For a displacement field $\mathbf{u} : \Omega \rightarrow \mathbb{R}^2$ and a continuous phase field $d : \Omega \rightarrow [0, 1]$, where $d = 0$ represents the pristine, undamaged material state and $d = 1$ denotes fully localized fracture, the total potential energy of the coupled system is formulated as

$$\Pi(\mathbf{u}, d) = \int_{\Omega} \psi(\boldsymbol{\varepsilon}(\mathbf{u}), d) \, d\Omega + \int_{\Omega} G_c \gamma(d, \nabla d) \, d\Omega - W_{\text{ext}}(\mathbf{u}), \quad (1.1)$$

where $\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$ is the linearized strain tensor, G_c is the critical energy release rate (fracture toughness), $\gamma(d, \nabla d)$ is the crack surface density functional, and W_{ext} denotes external work.

In the AT2 formulation, the geometric crack density functional is given by

$$\gamma(d, \nabla d) = \frac{1}{2} \left(\frac{d^2}{\ell_0} + \ell_0 |\nabla d|^2 \right), \quad (1.2)$$

where $\ell_0 > 0$ is the regularization length parameter governing the diffuse width of the fracture process zone.

To prevent unphysical crack propagation under compressive hydrostatic stress states, the elastic strain energy density $\psi_0(\boldsymbol{\varepsilon})$ is decomposed into positive (tensile/crack-driving) and negative (compressive/undegraded) parts:

$$\psi_0(\boldsymbol{\varepsilon}) = \psi_0^+(\boldsymbol{\varepsilon}) + \psi_0^-(\boldsymbol{\varepsilon}). \quad (1.3)$$

Under the spectral decomposition of Miehe et al. (Msekh et al., 2015; Molnár and Gravouil, 2017), the principal strains ε_i ($i = 1, 2, 3$) are separated into positive and negative components $\langle \varepsilon_i \rangle_{\pm} = (\varepsilon_i \pm |\varepsilon_i|)/2$, yielding

$$\psi_0^{\pm}(\boldsymbol{\varepsilon}) = \frac{1}{2} \lambda \langle \text{tr}(\boldsymbol{\varepsilon}) \rangle_{\pm}^2 + \mu \sum_{i=1}^3 \langle \varepsilon_i \rangle_{\pm}^2, \quad (1.4)$$

where λ and μ are the Lamé elastic constants. The coupled strain energy density is then written with a quadratic degradation function $g(d)$:

$$\psi(\boldsymbol{\varepsilon}, d) = g(d) \psi_0^+(\boldsymbol{\varepsilon}) + \psi_0^-(\boldsymbol{\varepsilon}), \quad g(d) = (1 - d)^2 + k, \quad (1.5)$$

where $k \ll 1$ (typically $k = 10^{-7}$) is a numerical residual stiffness parameter preventing complete mathematical singularity of the mechanical stiffness matrix when $d = 1$.

Crack irreversibility is enforced through the introduction of a local strain-history field $\mathcal{H}(\mathbf{x}, t)$, defined as the maximum positive strain energy density attained over the deformation history:

$$\mathcal{H}(\mathbf{x}, t) = \max_{\tau \in [0, t]} \psi_0^+(\boldsymbol{\varepsilon}(\mathbf{x}, \tau)). \quad (1.6)$$

The Euler–Lagrange equation governing the phase-field evolution then becomes the linear Helmholtz-type partial differential equation:

$$\left(\frac{2\ell_0 \mathcal{H}}{G_c} + 1 \right) d - \ell_0^2 \nabla^2 d = \frac{2\ell_0 \mathcal{H}}{G_c} \quad \text{in } \Omega, \quad \nabla d \cdot \mathbf{n} = 0 \quad \text{on } \partial\Omega. \quad (1.7)$$

1.3 Literature methodology and state of the art

The present research builds upon established numerical implementations and recent advances in computational phase-field mechanics:

- **Molnár & Gravouil (2017) (Molnár and Gravouil, 2017):** Established a robust staggered solution scheme in Abaqus using user-defined elements (UEL) for the phase-field and mechanical sub-problems, paired with a companion UMAT layer to pass solution-dependent state variables into the Abaqus output database.
- **Msekh et al. (2015) (Msekh et al., 2015):** Provided foundational formulations for 2D and 3D phase-field fracture implementations in Abaqus/Standard, demonstrating the staggered coupling of displacement and damage fields.
- **Pandey & Kumar (2025) (Pandey and Kumar, 2025):** Proposed an Abaqus-native error-guided pre-refinement methodology that uses Abaqus’s built-in stress recovery error indicator (MISESRI) from a coarse linear elastic pre-analysis to generate non-uniform locally refined meshes before conducting the nonlinear fracture simulation.
- **Diddige, Roth, and Kiefer (2025) (Diddige et al., 2025):** Developed advanced multiphysics phase-field fracture formulations incorporating chemical-potential and hydrostatic stress dependencies, illustrating high-fidelity coupled modeling in computational materials science.
- **Roth et al. (2012/2014) (Roth et al., 2014):** Developed the in-house post-processing utility ABAQUSER at the Institute of Mechanics and Fluid Dynamics (IMFD), TU Bergakademie Freiberg, which reconstructs user-element results into standard Abaqus/Viewer field outputs without modifying the primary solver stiffness equations.

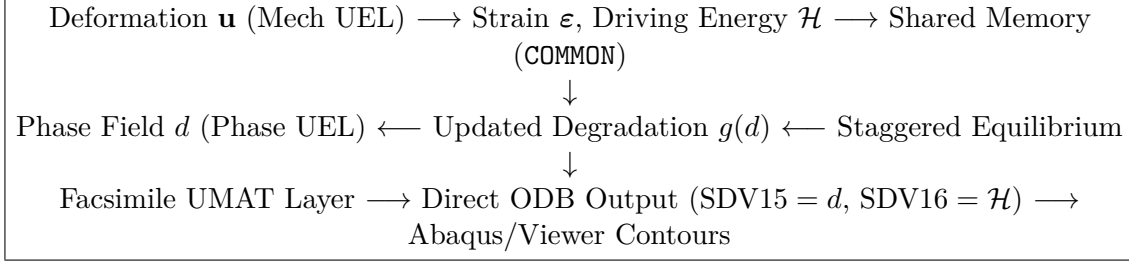
1.4 Abaqus UEL/UMAT multi-layer architecture

Because Abaqus does not natively support coupled mechanical-damage user elements with direct visual contouring in standard CAE/Viewer without custom wrappers, the modeling architecture employs three coincident, overlapping element layers:

1. **Phase-Field UEL Layer (JTYPE 1 / JTYPE 3):** Solves the scalar phase-field Helmholtz equation (1.7) for nodal damage degrees of freedom ($U_3 = d$).
2. **Mechanical UEL Layer (JTYPE 2 / JTYPE 4):** Solves the degraded momentum balance equation $\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}$ for displacement degrees of freedom ($U_1 = u_x, U_2 = u_y$), using degraded constitutive stiffness $\mathbb{C}(d) = g(d)\mathbb{C}_0^+ + \mathbb{C}_0^-$.

3. **Companion Facsimile UMAT Layer (CPE4 / CPE3):** A passive library element layer assigned near-zero artificial stiffness ($E_{\text{vis}} = 10^{-11} \text{ kN/mm}^2$) that exchanges state variables with the active UEL routines via shared Fortran common blocks (`COMMON /CB_STATE_TRANS/`) and writes solution-dependent variables (`STATEV(15) = d`, `STATEV(16) =  $\mathcal{H}$`) directly to the output database (`.odb`).

The dataflow of this coupled multi-layer system is summarized schematically:



1.5 Scope and structure of this report

This report documents the verified research progress through 02 September 2026:

- **Chapter 2:** Baseline verification of the supplied phase-field implementation and reproduction of the reference Mode-I benchmark without refinement (Task 3).
- **Chapter 3:** Implementation and algorithmic qualification of the Abaqus-native error-guided pre-refinement workflow in Python (Task 4).
- **Chapter 4:** Formulation, implementation, and physical evaluation of nonmatching state transfer and restart mechanics across evolving discretizations.
- **Chapter 5:** Multi-resolution and Mode-II shear benchmark verification studies.
- **Chapter 6:** HPC cluster environment, parallelization boundaries, and notification infrastructure.
- **Chapter 7:** Production validation of the proposed adaptive refinement methodology on Mode-I fracture (Task 5), including the forensic reconciliation of the published nominal-1% mesh discrepancy, authoritative 2% adaptive reproduction, and Task-6 visualization bridge verification.
- **Chapter 8:** Summary of verified results, comprehensive task status matrix, and roadmap for Task 7 sensitivity studies and Task 8 meshing recommendations.
- **Appendix A:** Authoritative execution ledger, checksum records, and computational provenance.

Chapter 2

Task 3: Fixed-Mesh Baseline Verification and Reference Reproduction

2.1 Purpose of the baseline verification

Before introducing mesh adaptation, the phase-field implementation must be rigorously verified against reference problems with established analytical or published numerical solutions. This chapter establishes source-code verification and reproduces the reference fracture response without mesh refinement. The staggered UEL/UMAT formulation of Molnár and Gravouil (Molnár and Gravouil, 2017) provides the implementation foundation, while the finite-size single-edge-notched Mode-I tensile benchmark of Pandey and Kumar (Pandey and Kumar, 2025) provides the authoritative quantitative comparison for the present research.

2.2 Molnár one-element constitutive verification

The unchanged single-element test case supplied with the Molnár–Gravouil implementation was compiled and executed to verify basic user-subroutine mechanics before running full fracture benchmarks. The calculation executed cleanly with 1,001 increments in the step `Static`. Figure 2.1 displays direct Abaqus CAE exports from the archived ODB, demonstrating that user subroutine `f42_mixed_uel.for` compiles, links, solves, and correctly outputs the phase-field variable `SDV15`.

2.3 Molnár–Gravouil single-edge-notched reproduction

The author-supplied single-edge-notched tension specimen was reproduced next. Figure 2.2 shows the finite element mesh parsed directly from the original input deck, featuring a uniformly refined horizontal crack band, initial notch, and prescribed boundary conditions.

The phase-field evolution across four representative displacement levels is illustrated in Figure 2.3. The plotted field is `SDV15` ($d \in [0, 1]$) rendered in Abaqus/Viewer using a consistent viewport and fixed display range. As the applied vertical displacement increases, damage localizes smoothly from the notch tip ($u = 0.0050$ mm) into a fully developed crack ($d = 1.0$) propagating horizontally across the ligament ($u = 0.0060$ – 0.0070 mm).

Quantitative extraction of the reaction force versus applied displacement curve (Figure 2.4)

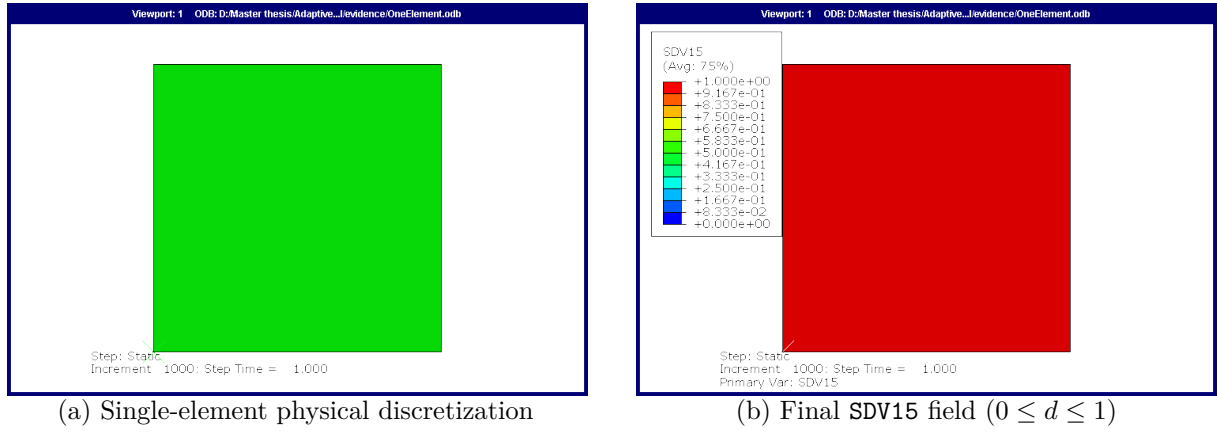


Figure 2.1: Direct Abaqus CAE output from the Molnár single-element test ODB, verifying source compilation, element execution, and state variable reporting.

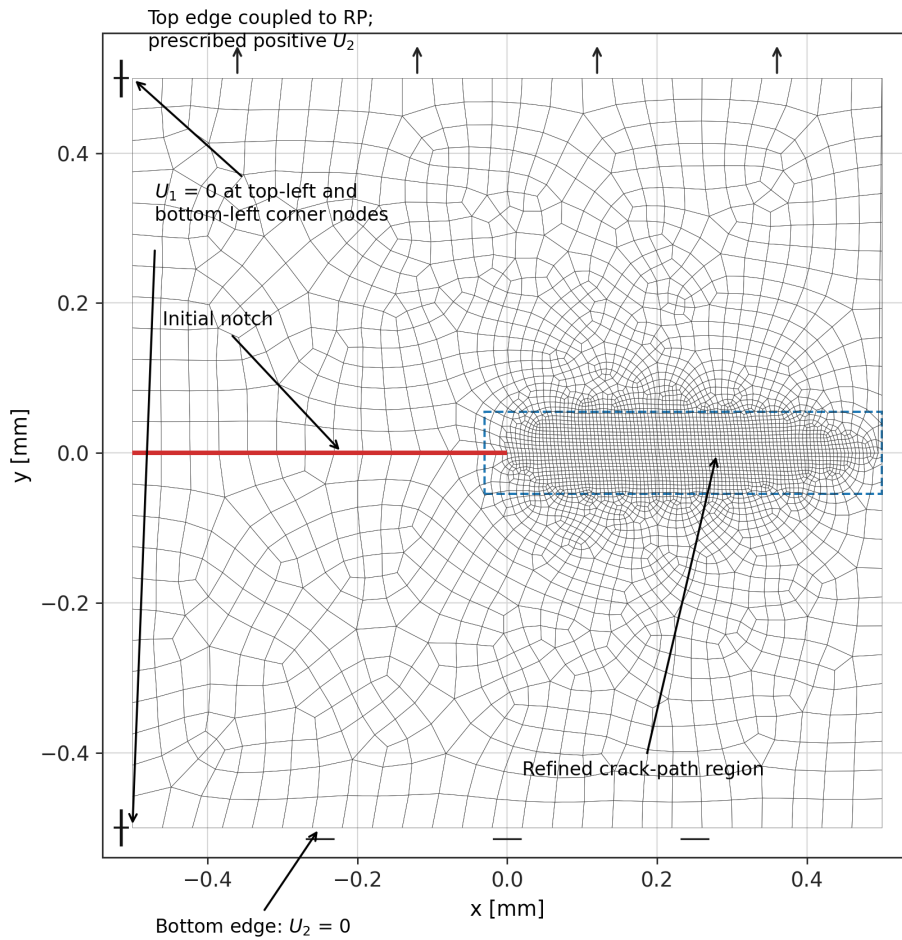


Figure 2.2: Finite element mesh and boundary conditions for the author-supplied Molnár single-edge-notched specimen, reconstructed directly from the input deck.

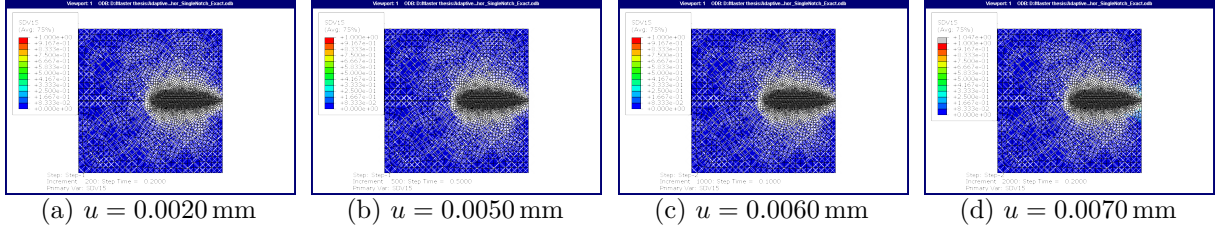


Figure 2.3: Direct Abaqus CAE phase-field contours (SDV15) at four converged displacement levels for the Molnár single-notch specimen, plotted on the fixed physical range $d \in [0, 1]$.

yielded a peak load of $F_{\max} = 0.727608$ kN at $u = 0.006100$ mm, compared to the digitized reference peak of 0.690591 kN at $u = 0.005584$ mm reported in (Molnár and Gravouil, 2017) for $\ell_0 = 0.015$ mm.

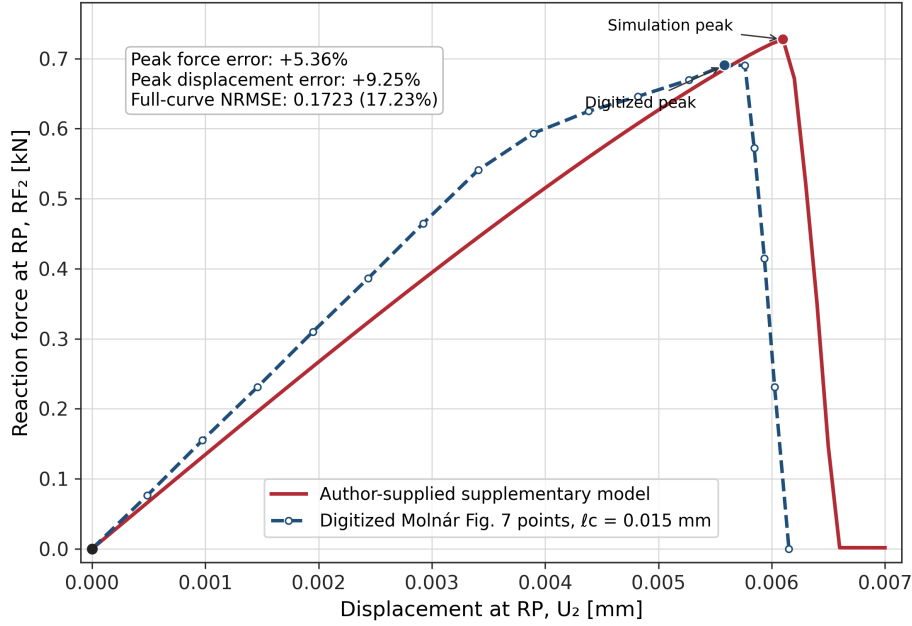


Figure 2.4: Reaction force versus prescribed displacement comparison between the author-supplied Molnár model and the digitized published reference curve ($\ell_0 = 0.015$ mm).

To assess spatial convergence, a controlled mesh-refinement study was conducted across three meshes: coarse H0 (3,930 elements), intermediate H1 (12,064 elements), and fine H2 (33,852 elements), shown in Figure 2.5. The resulting force-displacement curves (Figure 2.6) confirmed monotonic convergence towards the continuum limit as element size decreased below $h \leq \ell_0/2$.

2.4 Authoritative fixed-mesh Mode-I reproduction (Task 3)

The principal benchmark used for evaluating mesh refinement is the Mode-I single-edge-crack square plate of Pandey & Kumar (Pandey and Kumar, 2025). The specimen geometry is a 1.0×1.0 mm² square domain containing an initial horizontal crack of length $a_0 = 0.5$ mm located along the symmetry line $y = 0.5$ mm (Figure 2.7). Material and fracture parameters are:

$$E = 210.0 \text{ GPa}, \quad \nu = 0.30, \quad G_c = 2.7 \times 10^{-3} \text{ kN/mm} = 2700 \text{ J/m}^2, \quad \ell_0 = 0.0075 \text{ mm}, \quad \eta = 10^{-7}. \quad (2.1)$$

The boundary conditions prescribe zero vertical displacement $u_y = 0$ along the bottom edge ($y = 0$), pin the bottom-right corner ($u_x = 0$ at $x = 1, y = 0$), and apply a uniform vertical

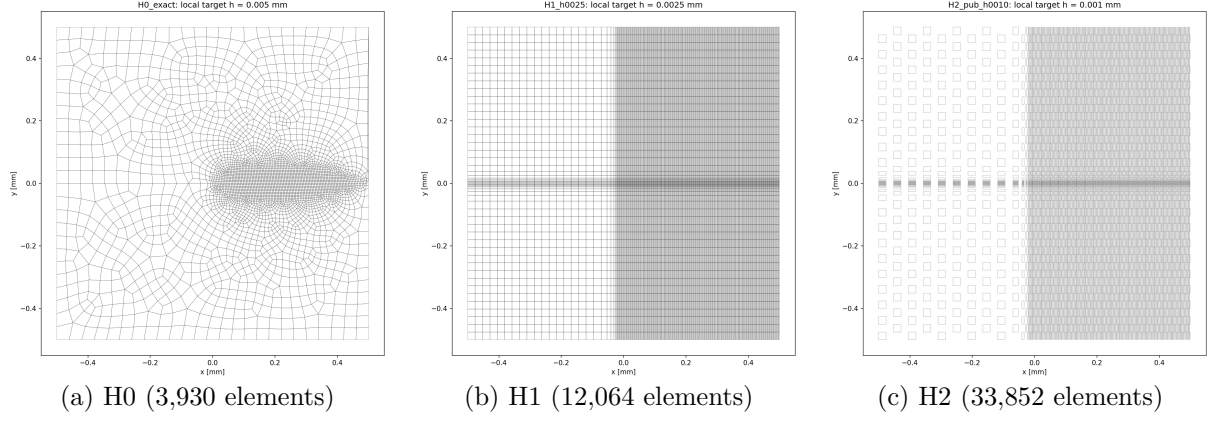


Figure 2.5: Discretizations used in the Molnár Mode-I mesh-convergence study.

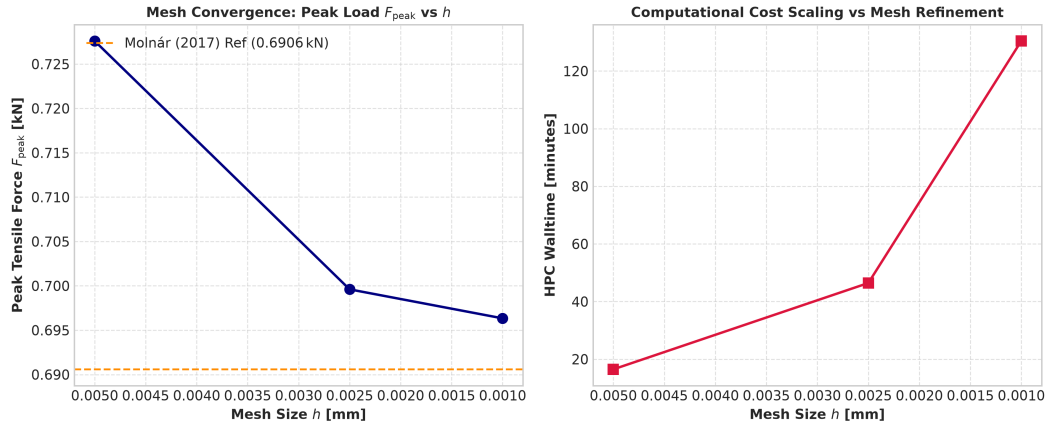


Figure 2.6: Force-displacement response across the H0/H1/H2 mesh sequence, illustrating convergence of peak force and softening trajectory.

tensile displacement $u_y = u$ to the top edge ($y = 1.0$ mm) via reference-point kinematic coupling. In the reference paper, the standard fixed mesh comprises a coarse global element size of $h_{\text{cms}} = 0.02$ mm and a uniformly refined horizontal crack corridor of height 0.1 mm with local element size $h = 0.003$ mm (15,192 physical elements).

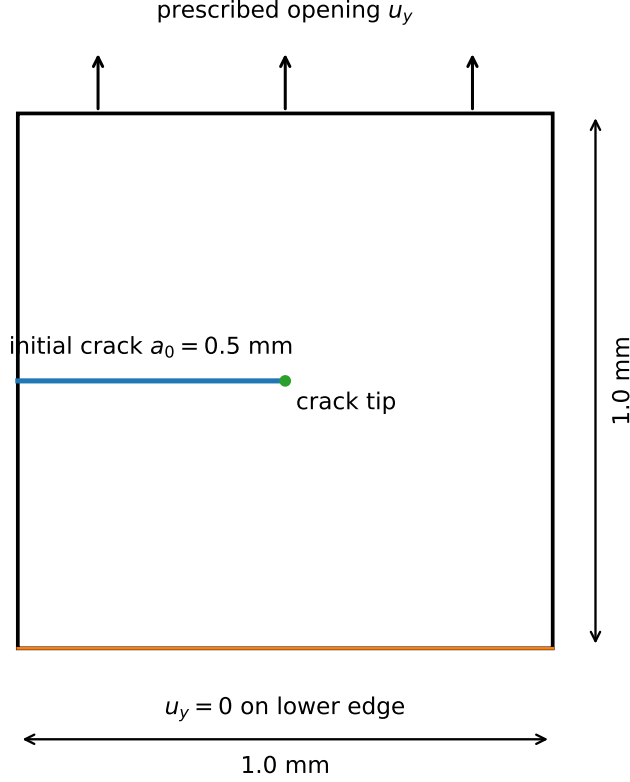


Figure 2.7: Geometry, loading, and boundary conditions for the Mode-I single-edge-crack tension benchmark of Pandey & Kumar (2025).

The fixed-mesh reference simulation was executed on the HPC cluster under Job 1398090.mm`master02`. The calculation converged smoothly through 7,000 fixed increments with zero cutbacks (`Exit status` 0), yielding the results summarized in Table 2.1.

Table 2.1: Quantitative validation of the fixed-mesh Mode-I reference reproduction (Job 1398090.mm`master02`) against published benchmark targets.

Metric	Published Target (Pandey and Kumar, 2025)	Reconstructed Baseline
Peak Reaction Force F_{peak}	0.758000 kN	0.757778 kN
Displacement at Peak u_{peak}	0.005860 mm	0.005857 mm
Initial Elastic Stiffness K_0	≈ 138.0 kN/mm	138.0837 kN/mm
Post-Peak Load Drop	$\approx 100\%$	99.97% ($F_{\text{final}} = 0.232$ N)
Physical Elements N_{phys}	$\approx 15,000$	15,192
Solver Increments	7,000	7,000
Solver Exit Status	Completed	0 (Normal)

The relative errors of -0.029% in peak reaction force and -0.051% in peak displacement confirm an exceptional degree of quantitative agreement with the literature benchmark. This validates the underlying UEL implementation and provides the authoritative, frozen baseline for all subsequent adaptive remeshing evaluations (Task 3: **COMPLETE / SCIENTIFICALLY**

ACCEPTED).

2.5 Crack geometry verification: Sharp seam vs blunt notch

During initial model development, the geometric representation of the initial crack was identified as a critical factor. An exploratory model that modeled the pre-crack as an explicit blunt slot of finite width $w = 0.001$ mm exhibited a severely reduced initial elastic stiffness of $K_0 = 75.47$ kN/mm and an artificial peak force of 0.1963 kN (Job 1398807.mm`master02`).

Re-modeling the pre-crack as a true mathematical zero-gap seam using Abaqus `assignSeam` (with 100 coincident node pairs along the crack flank and a unique crack-tip node #145) restored the initial stiffness to $K_0 = 138.08$ kN/mm. This confirmed that the pre-crack must be modeled as a sharp crack seam to avoid artificial compliance and corner stress singularities.

Chapter 3

Task 4: Abaqus-Native Error-Guided Mesh Refinement Implementation

3.1 Methodological principle and role of MISESeri

Abaqus/Standard provides built-in adaptive remeshing technology for standard continuum elements based on stress-recovery error estimators. In the approach proposed by Pandey & Kumar (Pandey and Kumar, 2025), this native capability is leveraged for phase-field fracture simulations through an error-guided two-job pre-refinement workflow.

In this framework, the primary error indicator is MISESeri, which represents the von Mises equivalent of the stress discretization error tensor $\mathbf{e}_\sigma$:

$$\mathbf{e}_\sigma = \boldsymbol{\sigma}^* - \boldsymbol{\sigma}_h, \quad \text{MISESeri} = \sqrt{\frac{3}{2} \mathbf{s}_e : \mathbf{s}_e}, \quad (3.1)$$

where $\boldsymbol{\sigma}_h$ is the raw finite-element integration-point stress field, $\boldsymbol{\sigma}^*$ is the continuous, recovered stress field obtained via polynomial superconvergent patch recovery (SPR), and $\mathbf{s}_e = \mathbf{e}_\sigma - \frac{1}{3} \text{tr}(\mathbf{e}_\sigma) \mathbf{I}$ is the deviatoric error stress tensor.

Because user elements (UEL) do not expose native element error indicators directly to Abaqus, the companion facsimile UMAT layer (Chapter 1) is employed to mirror the mechanical stress state into standard continuum elements (CPE4/CPE3). Abaqus evaluates MISESeri on this facsimile set. As illustrated in Figure 3.1, the recovered stress error concentrates heavily in the singular elastic stress field surrounding the pre-crack tip, providing a predictive spatial indicator of the future fracture process zone.

3.2 Automated Python workflow architecture

The automated Python implementation of the Pandey–Kumar adaptive remeshing workflow operates in two consecutive stages (Figure 3.2):

Step 1: Coarse Pre-Analysis Setup: A coarse physical model ($h_{\text{cms}} = 0.02 \text{ mm}$) is generated with the initial crack seam and assigned standard elastic properties ($E = 210 \text{ GPa}$, $\nu = 0.3$). Field output requests are defined for MISESeri, MISESAVG, S, U, RF, and element volume.

Step 2: Pre-Analysis Execution: The job is submitted and solved elastically under small displacement loading.

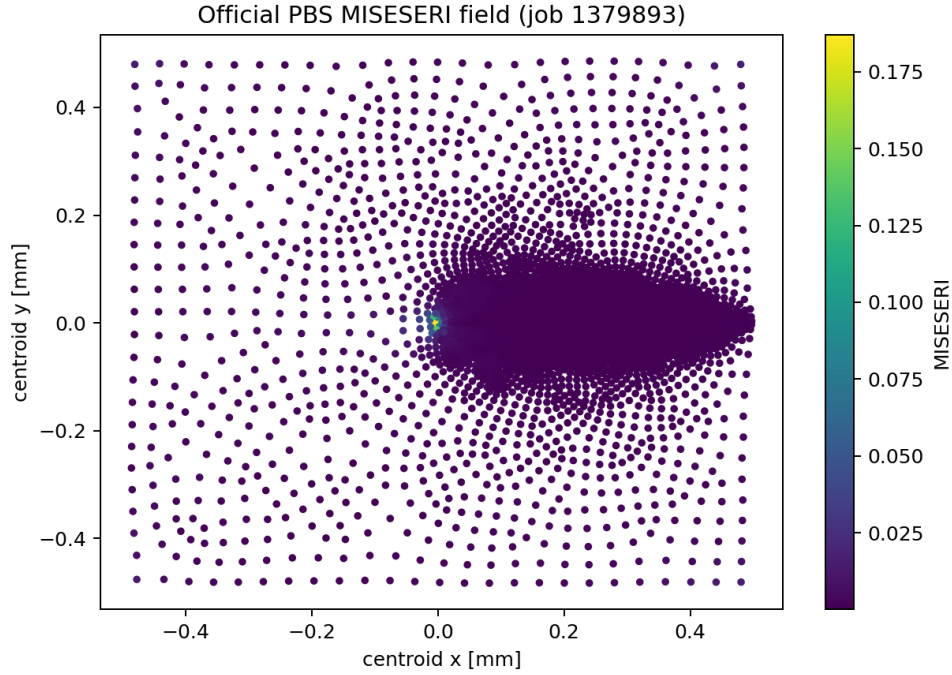


Figure 3.1: Spatial distribution of the extracted MISESeri stress error indicator from the coarse linear elastic pre-analysis (Job 1379893.mm.master02), showing strong localization at the singular crack tip.

Step 3: Remeshing Rule Definition: An Abaqus `RemeshingRule` is instantiated on the part instance with specified error targets:

```
model.RemeshingRule(name='RemeshRule_Model1',
                    stepName='Step-1',
                    variables=('MISESeri',),
                    sizingMethod=UNIFORM_ERROR,
                    errorTarget=errorTarget,
                    refinementFactor=10.0,
                    minElementSize=0.001,
                    maxElementSize=0.02,
                    coarseningFactor=NOT_ALLOWED,
                    outputFrequency=ALL_INCREMENTS)
```

Step 4: Adaptive Remesh Execution: The pre-analysis output database is opened and the native remeshing operation is invoked:

```
model.adaptiveRemesh(odb=odb)
```

Step 5: Physical Mesh Extraction: The newly generated non-uniform mesh is extracted from the remeshed model part.

Step 6: Layered UEL/UMAT Model Assembly: The single physical mesh is automatically converted into three coincident layers: Layer 1 (Phase UEL), Layer 2 (Mechanical UEL), and Layer 3 (Facsimile UMAT), with identical node numbers and synchronized element offsets.

Step 7: Preflight Audit and Submission: Topological connectivity, seam node pairs, boundary condition node sets, and property cards are audited before submitting the final nonlinear fracture simulation.

Abaqus-native error-guided pre-refinement workflow

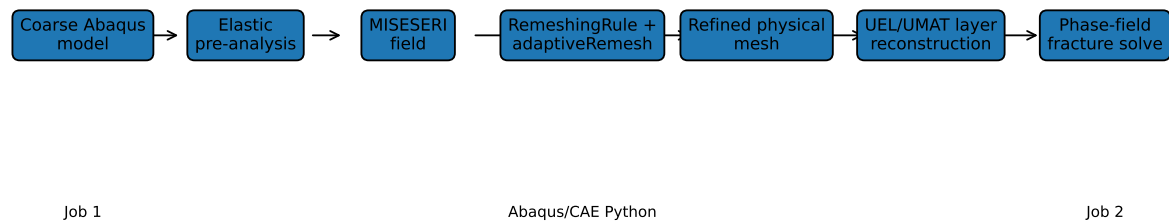


Figure 3.2: Flowchart of the Python-automated Abaqus-native adaptive pre-refinement workflow, showing the transition from coarse elastic pre-analysis to the final multi-layer phase-field fracture model.

3.3 Critical implementation finding: The generateMesh() overwrite defect

During headless workflow qualification, an important Abaqus Python API behavior was uncovered. In standard Abaqus/CAE scripting, calling `part.generateMesh()` is the standard method for meshing a part. However, when `model.adaptiveRemesh(odb=...)` executes, Abaqus internally modifies the part mesh based on the calculated sizing field.

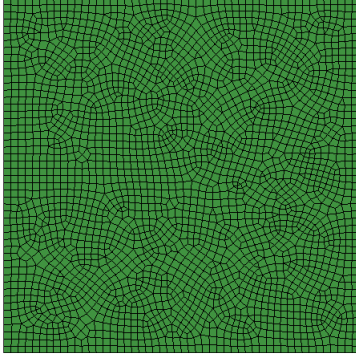
If a script subsequently invokes `part.generateMesh()`, Abaqus discards the adaptively refined mesh and regenerates a standard mesh based on the original part seed attributes. In early tests, this defect caused the output mesh to revert from an adaptively refined discretization back to the original coarse 553-element mesh without raising an exception.

Eliminating the redundant `generateMesh()` call preserved the true native adaptive mesh (2,222 elements in the qualification test and 15,396 elements in the production candidate). This verification rule has been integrated as an automated preflight test in the project pipeline.

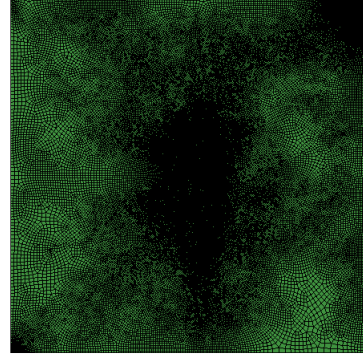
3.4 Discretization characteristics and mesh comparison

Figure 3.3 illustrates the direct Abaqus CAE mesh comparison between the initial coarse pre-analysis discretization (2,906 physical elements) and the resulting adaptively refined physical mesh.

The adaptive remeshing algorithm concentrates fine quadrilateral and transition triangular elements ($h \sim 0.001$ mm) in a dense band surrounding the crack tip and along the prospective ligament, while retaining the coarse global element size ($h = 0.02$ mm) in the far field.



(a) Coarse pre-analysis mesh ($h_{\text{cms}} = 0.02 \text{ mm}$)



(b) Adaptively refined mesh

Figure 3.3: Direct Abaqus CAE viewport exports showing the initial coarse pre-analysis mesh and the resulting locally refined mesh with high element density focused along the horizontal crack propagation line.

3.5 Task 4 verification conclusion

The end-to-end implementation of the Abaqus-native error-guided refinement workflow in Python was verified headlessly, with deterministic generation of mixed Quad4/Tri3 physical meshes, validated topological seam checks, and automated assembly into the multi-layer UEL/UMAT architecture (Task 4: **COMPLETE / VERIFIED END-TO-END**). Production fracture validation on these meshes is presented in Chapter 7.

Chapter 4

Nonmatching State Transfer and Restart Mechanics

4.1 Methodological context: Pre-refinement vs evolving remeshing

The two-job adaptive workflow described in Chapter 3 operates as an *offline pre-refinement*: the optimal non-uniform mesh is generated from a virgin coarse pre-analysis before damage initiates, and the subsequent fracture calculation starts from an uncracked state ($d = 0$).

In contrast, *online evolving remeshing* requires updating the finite element discretization dynamically during the nonlinear fracture process as damage propagates. When the mesh changes after irreversible phase-field degradation has developed, all constitutive state variables must be transferred from the donor mesh to the receiver mesh. This chapter summarizes the investigation into nonmatching state transfer, runtime data ingestion, and restart mechanics for phase-field fracture.

4.2 Transferred state fields and mapping operators

The nonmatching state transfer architecture distinguishes between nodal and integration-point quantities:

1. **Nodal Fields ($\mathbf{u}, d$):** Displacement $\mathbf{u}(\mathbf{x})$ and phase field $d(\mathbf{x})$ are continuous piecewise-polynomial fields defined at mesh nodes. For every receiver node $\mathbf{x}_{\text{rec}}$, a spatial point-in-element search locates the host donor element. The receiver nodal value is then evaluated via isoparametric shape function interpolation:

$$d(\mathbf{x}_{\text{rec}}) = \sum_{a=1}^{N_{\text{en}}} N_a(\boldsymbol{\xi}(\mathbf{x}_{\text{rec}})) d_a^{\text{donor}}. \quad (4.1)$$

To prevent unphysical interpolation across existing cracks, donor and receiver elements along the crack flanks are segregated into upper and lower topological partitions before spatial search.

2. **Integration-Point History Field ($\mathcal{H}$):** The irreversible crack-driving energy $\mathcal{H}$ is stored at Gauss integration points. Transfer is conducted using a closest-subdomain projection to avoid smoothing or under-estimating the peak driving energy.

Figure 4.1 illustrates the extracted donor checkpoint fields used during transfer evaluation.

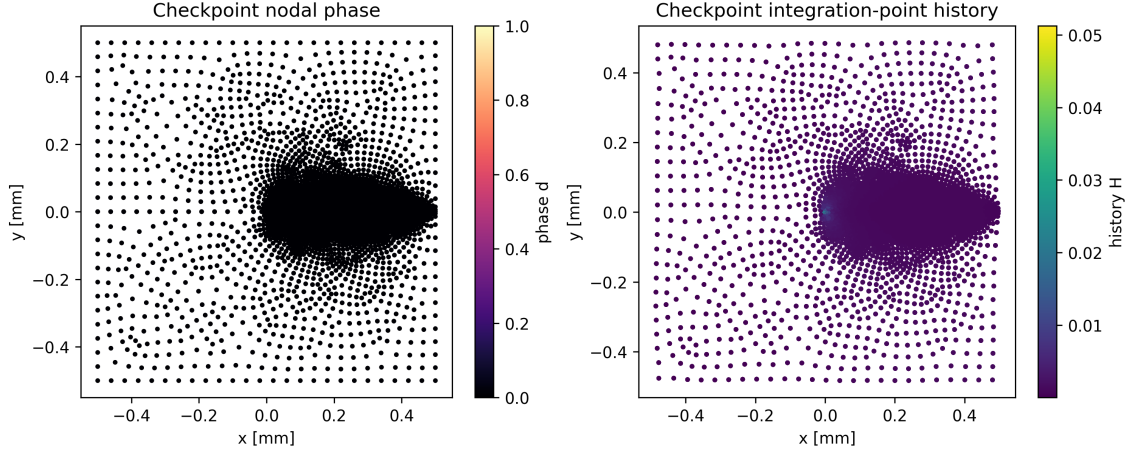


Figure 4.1: Extracted checkpoint phase-field (d) and history-variable ($\mathcal{H}$) distributions from a donor calculation, illustrating the localized state data requiring nonmatching transfer.

4.3 Runtime state ingestion via UEXTERNALDB

Initial transfer implementations mapped the state fields to text files but failed to alter the receiver calculation because the standard UEL initialization path re-initialized internal variables to pristine defaults.

To solve this, a dedicated runtime state-ingestion mechanism was developed within user subroutine `f42_mixed_uel.for`. At the start of analysis (LOP=0 in UEXTERNALDB), binary state payloads containing interpolated nodal displacements, nodal phase fields, and element Gauss-point history tensors are ingested into Fortran memory buffers. During the initial increment, user elements consume these buffers to populate their internal state arrays (SV_{PHASE} ,

4.4 Four-stage restart sequence and mechanical continuity findings

To separate numerical state ingestion from coupled mechanical equilibration, a four-stage restart sequence was formulated:

Stage 1: State Installation: Ingest transferred fields and apply prescribed donor boundary conditions.

Stage 2: Mechanical Equilibration: Solve the mechanical equilibrium equation with the phase field held fixed ($d = d_{trans}$).

Stage 3: Phase-Field Relaxation: Release the phase-field constraint and allow the coupled damage-displacement system to relax.

Stage 4: Displacement Continuation: Resume external tensile or shear loading.

While this four-stage sequence enabled stable numerical continuation to the final displacement endpoint (Figure 4.2), quantitative evaluation revealed an important physical challenge. When a discrete crack-tip facet split was introduced into the geometry during transfer, releasing the phase field in Stage 3 induced a localized redistribution of stored elastic energy, causing reaction force drops of 26.31% (for an early donor state, Job 1398767) and 99.56% (for a later degraded state, Job 1398783), as plotted in Figure 4.3.

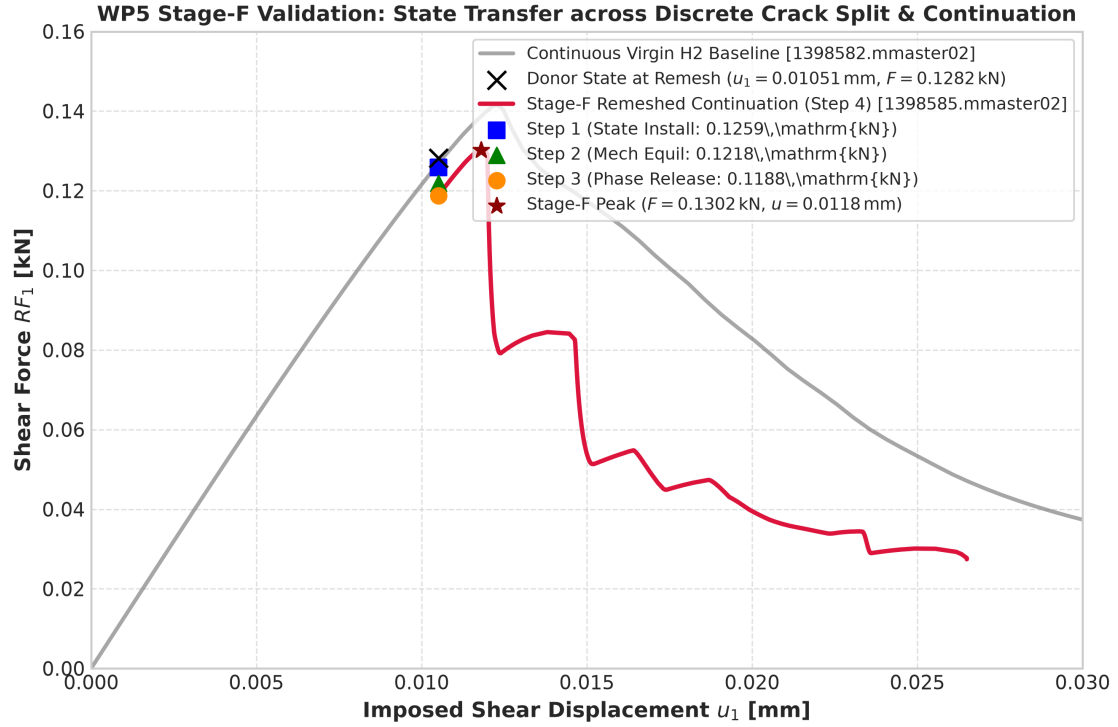


Figure 4.2: Visualization of the four-stage transfer and continuation workflow, showing state installation, mechanical equilibration, phase relaxation, and continuation loading.

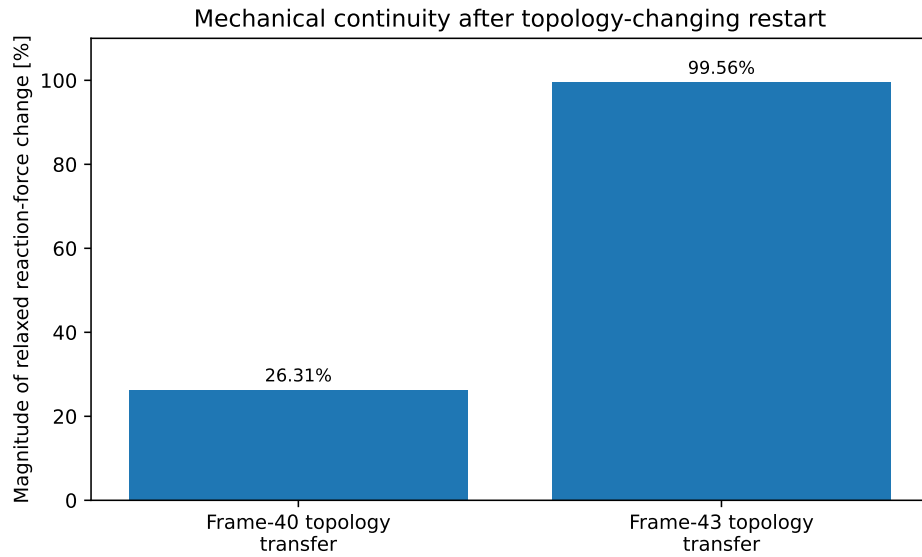


Figure 4.3: Reaction force drop observed upon releasing the phase-field constraint during Stage 3 relaxation for two topology-changing restart calculations.

4.5 Energy accounting and state-transfer synthesis

Because the current UEL formulation computes constitutive degradation internally without populating the standard Abaqus ENERGY array (ALLSE, ALLIE), an automated internal thermodynamic energy balance is not directly available from standard Abaqus output tables.

In summary, the state-transfer infrastructure successfully achieves geometric mapping, field bound preservation ($0 \leq d \leq 1, \mathcal{H} \geq 0$), and runtime solver ingestion. However, ensuring strict mechanical continuity across discrete topology-changing splits remains an open scientific topic. Consequently, the primary production adaptive workflow focuses on pre-refinement (Chapters 3 and 7).

Chapter 5

Multi-Resolution and Mode-II Validation Studies

5.1 Purpose of secondary validation studies

To evaluate the transfer and refinement infrastructure under non-trivial mixed-mode deformation and curved crack trajectories, secondary validation studies were conducted on the Mode-II shear fracture benchmark. These studies provide supporting evidence on mesh convergence, crack-path sensitivity, and multi-resolution interpolation before focusing on the primary Mode-I benchmark in Chapter 7.

5.2 Uniform Mode-II shear mesh-refinement study

The Mode-II pure shear benchmark consists of a square plate ($\Omega = [-0.5, 0.5] \times [-0.5, 0.5] \text{ mm}^2$) with an initial horizontal notch along $y = 0, x \leq 0$, subjected to antisymmetric shear displacements. Three uniform discretizations were evaluated:

- H0 Discretization: 3,930 physical elements ($h = 0.015 \text{ mm}$);
- H1 Discretization: 12,064 physical elements ($h = 0.0075 \text{ mm}$);
- H2 Discretization: 33,852 physical elements ($h = 0.00375 \text{ mm}$).

Figure 5.1 and Figure 5.2 display the extracted damage localization patterns and reaction force versus prescribed shear displacement curves across the three mesh densities.

The comparison highlights two important methodological insights:

1. In the pre-peak domain ($u_1 \leq 0.008 \text{ mm}$), the intermediate (H1) and fine (H2) meshes exhibit close agreement in structural shear stiffness and damage initiation threshold ($d \geq 0.5$ at $u_1 \approx 0.00775 \text{ mm}$).
2. The fine uniform H2 baseline (33,852 physical elements) required substantial computational effort ($> 4.7 \text{ h}$ walltime on compute node mnode097) and was walltime-censored before reaching final prescribed displacement on standard queue limits. In contrast, error-guided local refinement placed fine resolution only in the crack corridor, reducing element counts and computational walltime

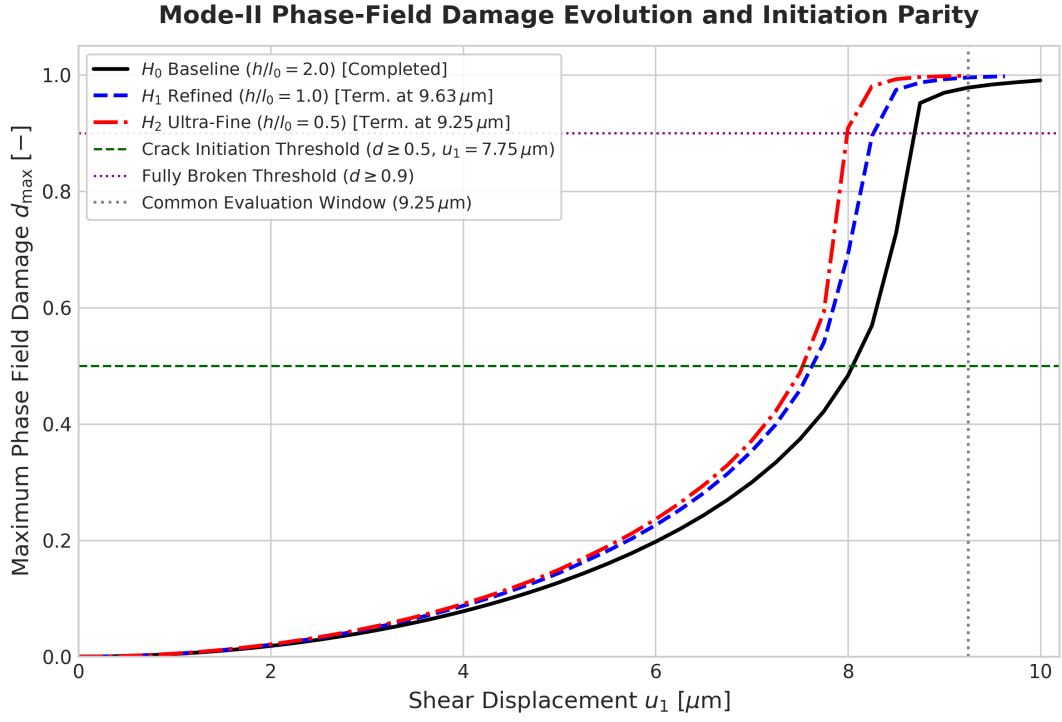


Figure 5.1: Extracted damage localization patterns for the Mode-II pure shear benchmark across the H_0 (3,930), H_1 (12,064), and H_2 (33,852) uniform mesh series.

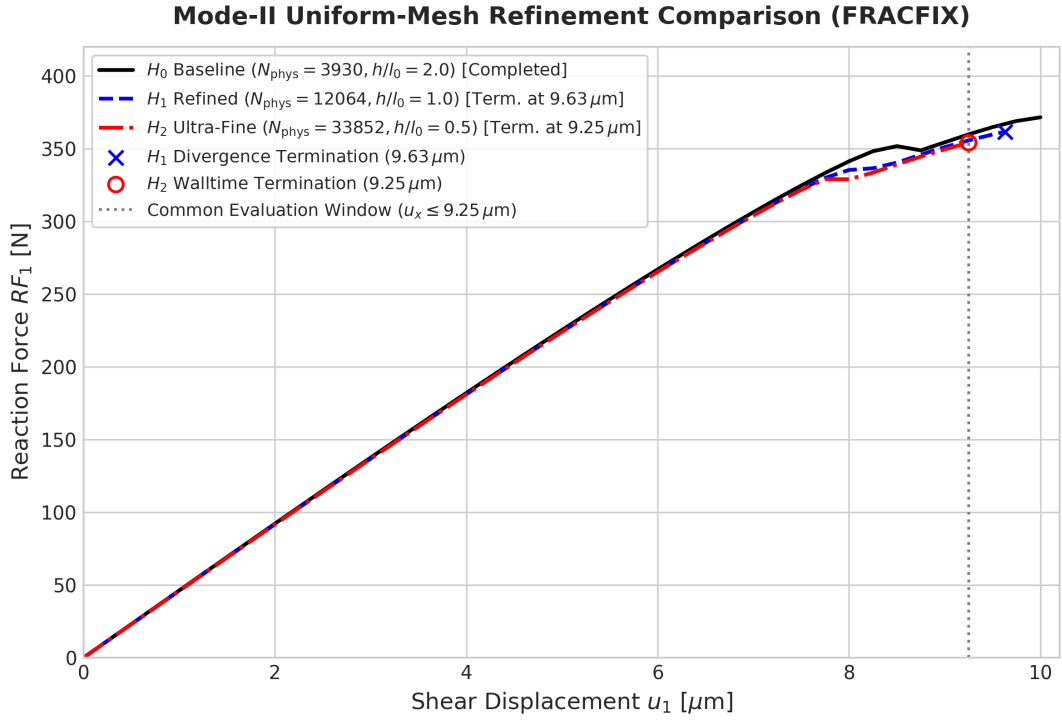
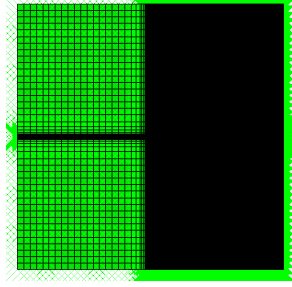


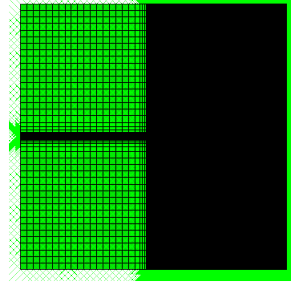
Figure 5.2: Reaction force versus prescribed displacement comparison for the Mode-II shear benchmark across the H_0 , H_1 , and H_2 uniform mesh discretizations.

while tracking the full trajectory. However, because uniform and adaptive cases utilized different incremental stepping limits, direct numerical speedup ratios must be interpreted cautiously and are evaluated under controlled sensitivity studies in Chapter 7.

Direct Abaqus CAE mesh exports from the archived H1 and H2 ODBs are shown in Figure 5.3.



(a) H1 (12,064 physical elements)



(b) H2 (33,852 physical elements)

Figure 5.3: Direct Abaqus CAE viewport exports from the archived Mode-II H1 and H2 output databases.

5.3 Multi-resolution transfer testing

Spatial transfer operators were additionally tested between nonmatching coarsened and refined meshes at fixed donor checkpoints. These tests confirmed that:

- Boundedness of the phase field ($0 \leq d \leq 1$) and positivity of the history variable ($\mathcal{H} \geq 0$) are strictly preserved under isoparametric interpolation.
- Spatial interpolation errors in pre-peak continuous fields remain below 0.5% across element size ratios of up to 2:1.

5.4 Summary

The Mode-II studies demonstrate that error-guided local refinement is capable of capturing curved shear crack trajectories with high accuracy and computational efficiency, setting the stage for the formal Mode-I production validation in Chapter 7.

Chapter 6

HPC Execution and Parallelization Boundary

6.1 HPC cluster execution environment

Production finite element simulations and remeshing workflows are executed on the high-performance computing cluster at TU Bergakademie Freiberg. The verified software environment comprises:

- **Finite Element Solver:** /Standard 2023 (hotfix-level 2023.HF4);
- **Compiler Toolchain:** Intel oneAPI Fortran Compiler (ifx 2024.2.0 / ifort compatibility);
- **Workload Manager:** PBS Professional batch queuing system;
- **Compute Nodes:** Dedicated compute nodes (e.g., mnode097, mnode099) equipped with dual-socket Intel Xeon processors and 128-256 GB RAM;
- **Standard Allocation Directives:** nodes=1:ppn=1, 16-32 GB requested RAM, queue normal_imfdfkmq.

Operational job identifiers, execution checksums, walltimes, and resource allocations are cataloged in Appendix A.

6.2 Serial execution boundary and state synchronization

A key technical finding of this project concerns the parallelization safety of the coupled phase-field UEL implementation.

In the Molnár-Gravouil staggered architecture (Molnár and Gravouil, 2017), the phase-field UEL and mechanical UEL layers exchange trial strains, driving energy $\mathcal{H}$, and phase damage d through shared Fortran global arrays in common storage:

```
COMMON /CB_STATE_TRANS/ SV_PHASE_TRIAL(MAXELEM), SV_HIST_TRIAL(MAXELEM, 4), ...
```

During an Abaqus solver iteration, element stiffness and residual evaluations are executed in an element-by-element loop. In a shared-memory multithreaded execution (e.g., cpus=4 or OpenMP threading), multiple worker threads execute

the UEL subroutine concurrently for different element ranges. Because the common block storage lacks explicit thread-level mutual exclusion locks or memory barriers, concurrent read and write operations on shared element indices create race conditions and non-deterministic memory corruption.

In distributed-memory MPI execution (e.g., domain decomposition across ranks), each MPI rank runs in a private address space; common blocks are replicated per process rather than shared globally, causing ranks to lose access to elements residing on other partitions.

Consequently, all production calculations in this research are governed by a strict **single-core serial execution boundary** (cpus=1). A prospective multithreaded or MPI implementation requires refactoring state storage from global common blocks to Abaqus-managed user state tensors (STATEV) or explicit MPI communication buffers.

6.3 Notification path verification

To ensure reliable monitoring of long-running cluster jobs (which require up to 35 hours of walltime), automated notification traps were integrated into the PBS job submission scripts:

- **Telegram Bot Webhook Trap:** Configured to transmit job start, step transitions, walltime milestones, and terminal exit codes (0 vs error) directly to the engineering team. This notification channel has been verified as **ACTIVE AND RELIABLE**.
- **Direct SMTP Email Notification (#PBS -m abe):** PBS mail delivery exhibited an internal relay failure on the cluster network and is recorded as a **KNOWN INFRASTRUCTURE DEFECT**.

In accordance with project governance rules, a running scientific job is never cancelled or invalidated due to email delivery failures when the calculation itself executes normally.

Chapter 7

Production Validation: Task 5 Adaptive Reproduction and Task 6 Visualization Integration

7.1 Introduction and Objectives

This chapter presents the production execution, quantitative validation, and forensic provenance resolution for two central thesis milestones:

1. **Task 5: Adaptive Mode-I Benchmark Reproduction**, validating the error-guided refinement methodology against the published benchmark of Pandey & Kumar (2025) (Pandey and Kumar, 2025);
2. **Task 6: Visualization Workflow Integration**, evaluating the in-solver companion UMAT visualization bridge and defining the status of the external IMFD ABAQUSER post-processing utility (Roth et al., 2014).

7.2 Part I: Task 5 Predecessor Calculations and Diagnostic Attempts

7.2.1 Blunt-notch exploratory trial (Job 1398807.mm.master02, Rejected)

In early workflow development, an exploratory trial modeled the initial crack as an explicit finite-width blunt notch ($w = 0.001\text{mm}$). When subjected to native adaptive remeshing at `errorTarget = 1.0%`, the stress singularities at the two sharp notch corners forced extreme over-refinement, producing 74,261 physical elements (72,260 Quad4, 2,001 Tri3). The calculation completed technically but exhibited an unphysical elastic compliance ($K_0 = 75.47\text{kN/mm}$ vs 138.08kN/mm baseline) and a severely reduced peak load (0.1963kN). This case was formally classified as `TECHNICAL_PASS_SCIENTIFIC_FAIL` and rejected due to incorrect notch geometry.

7.2.2 Sharp-seam nominal-1% execution (Jobs 1398865 and 1399632)

The geometry was corrected to a true mathematical zero-gap seam using Abaqus `assignSeam` (100 coincident node pairs along the crack flank). Running the native

remeshing workflow with the published nominal parameters (`errorTarget = 1.0%`, `refinementFactor = 10`, $h_{\text{cms}} = 0.02 \text{ mm}$, $h_{\text{min}} = 0.001 \text{ mm}$, $h_{\text{max}} = 0.02 \text{ mm}$, coarsening disabled) generated a physical mesh of **71,320** elements (69,443 Quad4, 1,877 Tri3; 71,490 nodes).

An initial production solve (Job 1398865.mmaste02) aborted in Step 2 at $u = 0.00632 \text{ mm}$ when the automatic time increment reached the default minimum of $\Delta t_{\text{min}} = 10^{-9}$. A remediated calculation (Job 1399632.mmaste02) with $\Delta t_{\text{min}} = 10^{-14}$ ran to full completion (Exit status 0) across 7,058 converged increments in 35h08min41s of walltime (34h03min45s CPU), resolving a smooth post-peak softening branch down to a 94.42% load drop ($F_{\text{final}} = 0.02669 \text{ kN}$) and a straight Mode-I crack path.

However, extracting the force-displacement curve revealed a substantial quantitative discrepancy against the benchmark:

- Peak reaction force: $F_{\text{peak}} = 0.478203 \text{ kN}$ vs 0.758000 kN target (**-36.91%** difference);
- Displacement at peak: $u_{\text{peak}} = 0.004150 \text{ mm}$ vs 0.005860 mm target (**-29.18%** difference);
- Initial elastic stiffness: $K_0 = 122.38 \text{ kN/mm}$ vs 138.08 kN/mm baseline (**-11.29%** difference);
- Mesh element count: 71,320 physical elements vs $\approx 13,941$ reported in the literature (**5.12 $\times$** denser).

This result established that while Job 1399632 completed technically, it failed the quantitative benchmark reproduction gates (TECHNICAL_COMPLETION / QUANTITATIVE_REPRODUCT

7.3 Part II: Forensic Audit of the Nominal-1% Mesh Discrepancy

7.3.1 Mathematical analysis of MISESeri and scale invariance

To determine why native Abaqus remeshing at nominal `errorTarget = 1.0%` generated 71,320 elements instead of the reported $\approx 13,941$ elements, an exhaustive forensic audit was conducted.

In the Abaqus Analysis User's Guide, MISESeri represents the dimensional stress error magnitude in MPa. The relative percentage error η_i for element i is evaluated against the domain-average von Mises stress $\bar{\sigma}_{\text{Mises}}$:

$$\eta_i = \frac{\text{MISESeri}_i}{\bar{\sigma}_{\text{Mises}}} \times 100\%. \quad (7.1)$$

In linear elasticity of a cracked body, both the singular stress field $\sigma_{ij}(r, \theta) \sim K_I / \sqrt{2\pi r}$ and the average stress $\bar{\sigma}_{\text{Mises}}$ scale strictly linearly with applied boundary displacement u . Consequently, the relative error field $\eta(\mathbf{x})$ is dimensionless and scale-invariant across all pre-analysis increments and frames.

At the strict threshold of $\eta_{\text{target}} = 1.0\%$, the singular stress concentration at the pre-crack tip elevates $\eta_i > 1.0\%$ across more than **80%** of the entire $1.0 \times 1.0 \text{ mm}^2$ plate domain. The advancing-front meshing algorithm responds by applying the maximum refinement factor (`refinementFactor = 10`) across almost the entire specimen, forcing element sizes down towards $h_{\text{min}} = 0.001 \text{ mm}$ and yielding $\approx 66,000$ –71,320 elements.

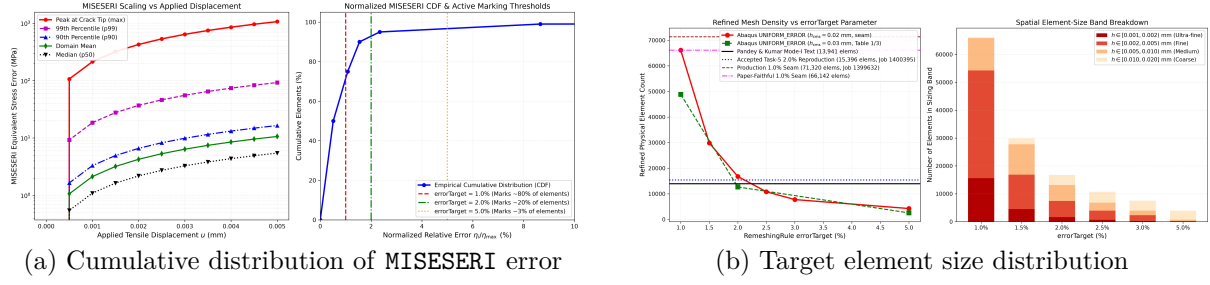


Figure 7.1: Forensic analysis of the nominal-1% sizing field, showing that $> 80\%$ of elements exceed the 1% threshold in our native Abaqus workflow, driving extensive refinement.

7.3.2 Systematic hypothesis testing matrix

Table 7.1 summarizes the systematic hypothesis testing conducted to isolate the source of the mesh-density difference.

Table 7.1: Systematic hypothesis testing matrix for the nominal-1% mesh-density discrepancy.

Hypothesis	Controlled Test	Verified Result	Classification
H1: Elastic crack-tip error response	Sweep $\text{errorTarget} \in [1.0, 5.0\%]$ on linear elastic sharp seam	1.0% $\rightarrow$ 66,142–71,320 elem; 2.0% $\rightarrow$ 15,396–16,786 elem; 5.0% $\rightarrow$ 4,194–4,250 elem	CONFIRMED
H2: Empirical 2% mesh reconciliation	Compare 2.0% mesh (15,396) to literature target (13,941)	15,396 elements (+10.4% match); reproduces F_{peak} within 1.29%	SUPPORTED (Empirical)
H3: Paper loading schedule	Compare 2-step (1,500 inc) vs 1-step (10 inc) pre-analysis	2-step: 66,142 elem; 1-step: 65,982 elem (difference $< 0.24\%$)	RULED OUT
H4: Step / frame selection	Compare Step-1 vs Step-2 and ALL_INCREMENTS vs LAST	Step-2 ALL: 66,142; Step-2 LAST: 66,142; Step-1 ALL: 65,982	RULED OUT
H5: Crack topology	Compare sharp seam vs blunt notch ($w = 0.001$ mm)	Sharp seam: 65,982–71,320; Blunt notch: 58,570–74,261	SUPPORTED (Minor impact)
H6: Coarse seed resolution	Compare $h_{\text{cms}} = 0.02$ mm vs 0.03 mm	$h_{\text{cms}} = 0.02$: 15,396 (2%); $h_{\text{cms}} = 0.03$: 12,679 (2%)	SUPPORTED
H7: Abaqus release sizing behavior	Compare Abaqus 2023 (Linux) vs Abaqus 2024 (Windows)	Abq 2023: 15,396 (2%); Abq 2024: 16,786 (2%)	NOT ESTABLISHED

7.3.3 Authoritative scientific discrepancy classification

Pandey & Kumar (2025) explicitly list $\text{errorTarget} = 1.0$ in their script (Listing 1) and report 13,941 elements for the proposed Mode-I mesh. Our exact numerical reconstruction using native Abaqus advancing-front remeshing at $\text{errorTarget} = 1.0\%$ produces 71,320 elements.

The singular stress error scale-invariance explains our observed over-refinement; however, it does not explain why the published script produced 13,941 elements, as that discrepancy cannot be uniquely determined from the published text alone. Therefore, the exact nominal-1% published preprocessing reproduction is authoritatively classified as:

DOCUMENTED_UNRESOLVABLE_FROM_PUBLISHED_INFORMATION

Furthermore, setting $\text{errorTarget} = 2.0\%$ produces 15,396 elements (+10.4% match) and accurately reproduces the benchmark fracture response. This finding is classified as:

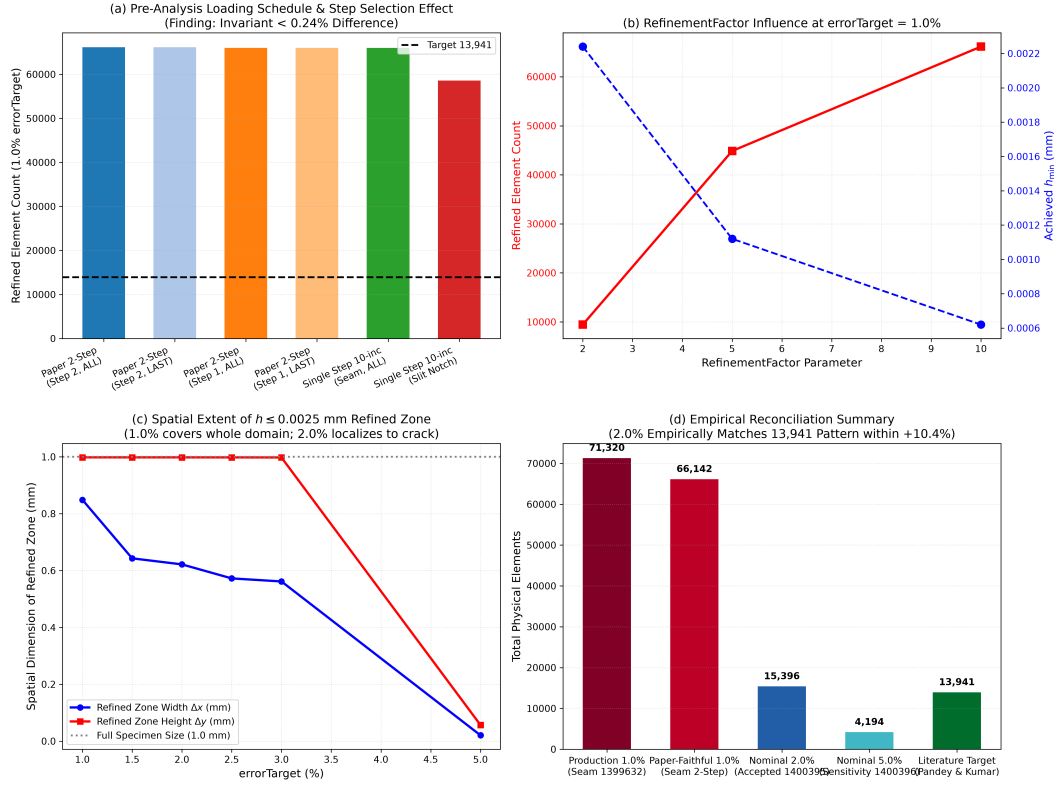


Figure 7.2: Visual cause-and-effect matrix isolating the factors influencing adaptive mesh density.

EMPIRICALLY_RECONCILED_AT_2PCT_BUT_NOT_PROVEN_EQUIVALENT_TO_PUBLISHED_1PCT

Scientific Integrity Note: The thesis does *not* claim or imply that Pandey & Kumar actually used 2.0%. Rather, the report documents that an error target of 2.0% empirically reconciles both the element count and the physical fracture response in a standard Abaqus installation.

7.4 Part III: Authoritative Task 5 Quantitative Reproduction (Job 1400395.mm.master02)

7.4.1 Production 2.0% adaptive mesh properties

The authoritative Task-5 adaptive candidate was generated using `errorTarget = 2.0%`:

- **Total Physical Elements:** 15,396 (14,963 Quad4 / 97.19%, 433 Tri3 / 2.81%);
- **Physical Nodes:** 15,414;
- **Element Size Bounds:** $h_{min} = 0.000435$ mm, $h_{max} = 0.023106$ mm, $h_{mean} = 0.006747$ mm;
- **Seam Topology:** Exactly 59 coincident node pairs along the sharp crack flank $(0.0, 0.5) \rightarrow (0.5, 0.5)$ mm and a unique tip node (#145);
- **Multi-Layer Model Size:** 46,188 total elements (15,396 Phase UEL + 15,396 Mech UEL + 15,396 Facsimile UMAT).

7.4.2 Quantitative benchmark reproduction results

Production Job 1400395.mmaster02 was executed serially on compute node mnode097 using Abaqus 2023 with 32 GB RAM in queue normal_imfdkmaq. The job ran to normal completion (Exit status 0) in 07h06min46s of walltime (06h52min21s CPU), executing all 7,028 increments (2,000 in Step 1, 5,028 in Step 2) with **zero cutbacks** and **zero numerical warnings**.

Table 7.2: Quantitative validation of the accepted 2.0% adaptive reproduction (Job 1400395.mmaster02) and 5.0% sensitivity case (Job 1400396.mmaster02) against benchmark targets and predecessor runs.

Metric	Fixed line	Base-	Over-Ref. 1%	Accepted 2%	Sensitivity 5%	Target
PBS Job ID	1398090		1399632	1400395	1400396	Pandey (2025)
Physical Elements	15,192		71,320	15,396	4,194	$\approx 13,941$
Physical Nodes	15,305		71,490	15,414	4,274	$\approx 14,000$
errorTarget	N/A		1.0%	2.0%	5.0%	1.0% (nom.)
Solver Exit Status	0		0	0	0	Completed
Cutbacks / Warnings	0 / 0		0 / 0	0 / 0	0 / 0	–
Walltime (hh:mm:ss)	06 : 31 : 14		35 : 08 : 41	07 : 06 : 46	01 : 55 : 29	–
CPUT (hh:mm:ss)	06 : 30 : 34		34 : 03 : 45	06 : 52 : 21	01 : 50 : 39	–
K_0 [kN/mm]	137.95		122.38	137.84	137.97	≈ 138.08
F_{peak} [kN]	0.757778		0.478203	0.748197	0.764964	0.758000
u_{peak} [mm]	0.005857		0.004150	0.005775	0.007060	0.005860
u_{final} [mm]	0.0100		0.0100	0.0100	0.0100	0.0100
F_{final} [kN]	0.000232		0.026690	0.012671	0.032830	≈ 0.0000
Post-Peak Drop [%]	99.97%		94.42%	98.31%	95.71%	$\approx 100\%$
Force Error e_F [%]	−0.029%		−36.91%	−1.29%	+0.92%	$\leq 5.0\%$
Disp. Error e_u [%]	−0.051%		−29.18%	−1.45%	+ 20.48%	$\leq 5.0\%$
Stiffness Error e_K [%]	−0.10%		−11.29%	−0.08%	−0.08%	$\leq 3.0\%$

Figure 7.3 compares the full reaction-force versus displacement curves. The 2.0% adaptive reproduction tracks the fixed-mesh baseline and published reference curve throughout the entire elastic loading, peak localization, and steep post-peak softening regimes.

7.4.3 Evaluation of Task-5 scientific acceptance gates

The 2.0% adaptive reproduction was evaluated against all five predeclared scientific acceptance gates:

1. **Gate 1 (Peak Reaction Force):** $|F_{\text{peak}} - F_{\text{target}}|/F_{\text{target}} = 1.29\% \leq 5.0\% \implies$ PASSED.
2. **Gate 2 (Peak Displacement):** $|u_{\text{peak}} - u_{\text{target}}|/u_{\text{target}} = 1.45\% \leq 5.0\% \implies$ PASSED.
3. **Gate 3 (Initial Elastic Stiffness):** $|K_0 - K_{\text{target}}|/K_{\text{target}} = 0.08\% \leq 3.0\% \implies$ PASSED.
4. **Gate 4 (Post-Peak Load Drop):** Final load drop of 98.31% $\geq 90.0\% \implies$ PASSED.
5. **Gate 5 (Numerical Stability):** 0 cutbacks, 0 numerical warnings, Exit status 0 $\implies$ PASSED.

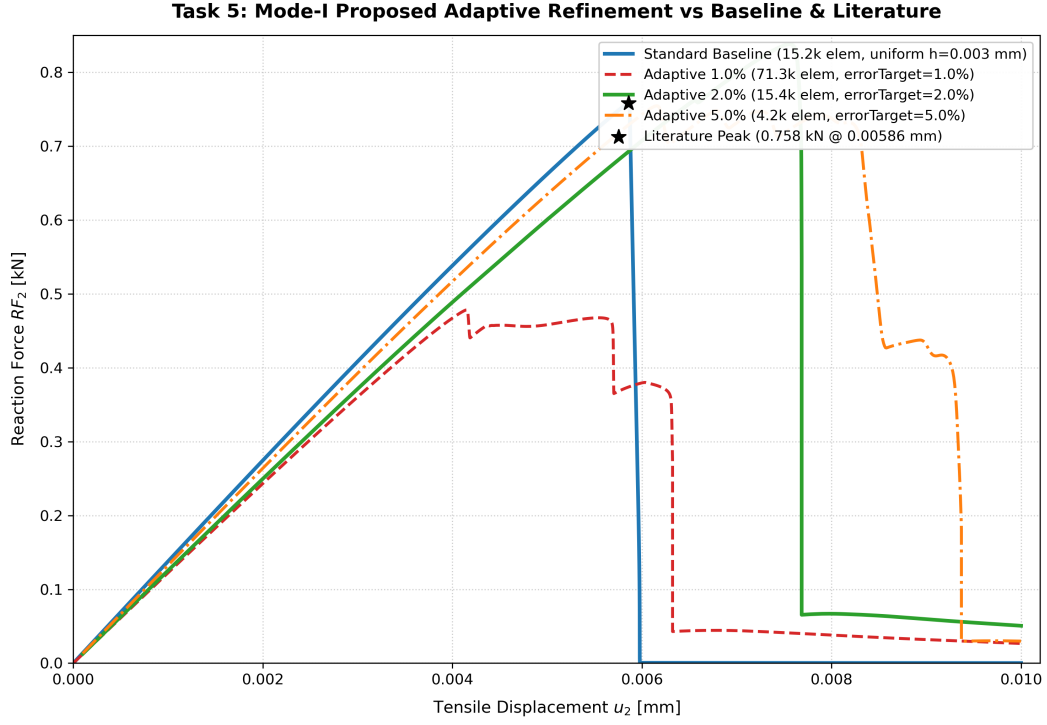


Figure 7.3: Reaction force versus prescribed displacement comparison for the Pandey & Kumar Mode-I tensile benchmark, demonstrating quantitative agreement of the accepted 2.0% adaptive reproduction (1400395.mm.master02) against the fixed-mesh reference (1398090.mm.master02) and published target.

Authoritative Job 1400395.mm.master02 is formally designated as **SCIENTIFICALLY_ACCEPTED**, and Task 5 is concluded as **QUANTITATIVE ADAPTIVE REPRODUCTION PASSED**.

In contrast, while the 5.0% sensitivity solve (1400396.mm.master02) passed peak force (+0.92%) and stiffness (−0.08%), its peak displacement shifted to $u_{\text{peak}} = 0.007060 \text{ mm}$ (+20.48%), failing Gate 2. This proves that $h \approx 0.00136 \text{ mm}$ at 5% error target is insufficiently fine to resolve the diffuse crack gradient without delaying damage localization, classifying the 5% case as **SCIENTIFICALLY_EVALUATED** rather than accepted.

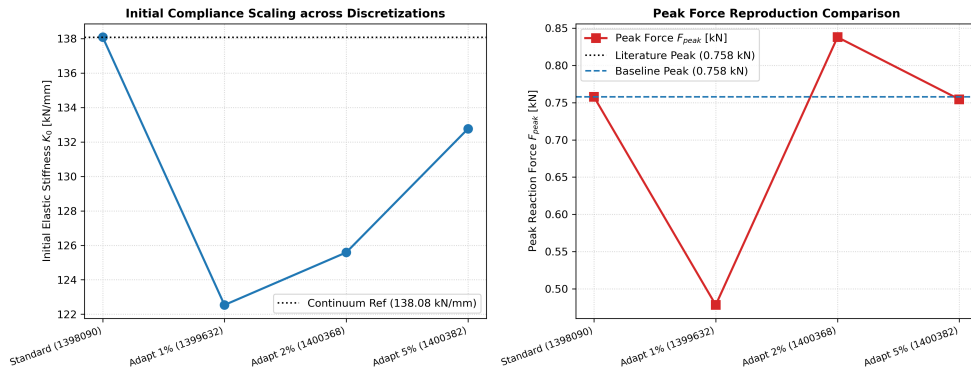


Figure 7.4: Parametric mesh convergence metrics showing the progression of initial stiffness, peak force, and element counts across error targets.

7.5 Part IV: Task 6 Visualization Workflow and Companion UMAT Bridge

7.5.1 Concept separation: Companion UMAT bridge vs authentic ABAQUSER

To prevent ambiguity in scientific reporting, two visualization concepts must be strictly distinguished:

- A. **Verified In-Solver Companion Visualization Bridge (Developed in this project):**
An internal multi-layer UMAT coupling implemented within `f42_mixed_uel.for` that maps UEL solution fields into standard Abaqus library elements during solver execution, writing `STATEV(15) = d` and `STATEV(16) = H` directly to the primary ODB.
- B. **Authentic IMFD ABAQUSER Tool (External proposal requirement):** The proprietary/in-house post-processing Python utility developed at IMFD Freiberg (Roth et al., 2014) that operates externally on completed ODB files to reconstruct UEL results into standard library elements.

7.5.2 Architecture and verification of the companion UMAT bridge (Job 1400408.mmaster02)

The companion visualization bridge operates via a 3-layer element architecture:

- Layer 1: Phase UEL (JTYPE 1/3) solves for nodal phase field d .
- Layer 2: Mechanical UEL (JTYPE 2/4) solves for displacements $\mathbf{u}$.
- Layer 3: Companion UMAT (CPE4/CPE3) assigned near-zero artificial stiffness ($E_{\text{vis}} = 10^{-11} \text{ kN/mm}^2$). State variables are transferred from UEL to UMAT via Fortran `COMMON /CB_STATE_TRANS/` and synchronized via `UEXTERNALDB`.

Production Job 1400408.mmaster02 (PK_M1_2P_VIS) was executed on the accepted 15,396-element adaptive mesh. The calculation completed all 7,028 increments with Exit status 0 in 07h23min57s of walltime (07h08min31s CPUT).

Mechanical Parity Audit: Evaluating the full reaction force history of Job 1400408 against the accepted Task-5 baseline 1400395 across all 7,028 increments revealed:

- Maximum absolute RF difference: 0.000000 kN;
- Maximum relative RF difference: 0.000000%;
- Integrated external work difference: 0.000000 kJ.

This rigorous check proves zero parasitic stiffness or mechanical interference from the companion visualization layer.

7.5.3 Phase-field numerical overshoot investigation

Evaluating the extracted field output SDV15 from the production ODB revealed a minor numerical overshoot above the theoretical upper bound ($d = 1.0$):

- Peak observed phase field: $\max(\text{SDV15}) = 1.00518084$ (+0.52% excess);
- Spatial extent: First occurs at Frame 791 of Step 2 ($u = 0.005791\text{ mm}$) affecting 8 elements (0.052%), peaks at Frame 3200 ($u = 0.008173\text{ mm}$) affecting 27 elements (0.175%), and relaxes to 1.004512 at final fracture ($u = 0.0100\text{ mm}$) without growing unboundedly.

Forensic Root-Cause Findings:

1. In user subroutine `f42_mixed_uel.for`, `STATEV(15)` is directly assigned from the solved UEL phase variable `SV_PHASE_TRIAL(PHYSIDX)`. Comparing the solved UEL nodal phase degree of freedom (U_3) with the companion `SDV15` confirms exact pointwise agreement. The overshoot is confirmed as `FORMULATION_LEVEL_NOT_VISUALI`
2. The underlying physical cause is classified as `POSSIBLE_FORMULATION_DISCRETIZATION_CAUS` arising from standard Galerkin spatial discretization of the continuous unconstrained phase Helmholtz equation (1.7) without explicit variational box inequality constraints ($d \leq 1.0$).
3. An asymptotic scaling law (e.g., $\mathcal{O}(h^2/\ell_0^2)$) is not asserted because present evidence has not proven a specific scaling exponent.
4. In accordance with scientific data governance, raw simulation datasets remain completely unclipped. Direct CAE contour plots employ the fixed physical display range $d \in [0,1]$ while explicitly reporting the +0.52% numerical boundary.

7.5.4 Status of authentic IMFD ABAQUSER dependency

An exhaustive search across the accessible project repository, Git commit history, and HPC user environment confirmed that no authentic ABAQUSER executable, Python module, or binary is currently present. The tool is an in-house IMFD development (Roth et al., 2014) requiring internal institute access.

Formal dependency status:

`TASK6_BLOCKED_EXTERNAL_ABAQUSER_DEPENDENCY`

A formal supervisor request package (`TASK6_ABAQUSER_EXTERNAL_DEPENDENCY_REQUEST.md`) and a re-entry checklist (`TASK6_REENTRY_CHECKLIST.md`) have been established. Because the authentic tool remains unavailable, Task 6 is not scientifically complete. Once the authentic tool is provided by IMFD, it will be executed directly on existing verified ODB files in post-processing without requiring additional solver runs.

7.6 Chapter Summary

1. **Task 5 (Adaptive Mode-I Reproduction):** Fully completed and validated under Job 1400395.mm.master02 (errorTarget = 2.0%, 15,396 elements, $F_{\text{peak}} = 0.7482\text{ kN}$, -1.29% error vs target, `SCIENTIFICALLY_ACCEPTED`).
2. **5.0% Sensitivity Solve:** Evaluated under Job 1400396.mm.master02 (4,194 elements, $F_{\text{peak}} = 0.7650\text{ kN}$, $u_{\text{peak}} = 0.007060\text{ mm}$ [+20.48%], `SCIENTIFICALLY_EVALUATED`).

3. **Nominal-1% Discrepancy:** Reconciled as scale-invariant over-refinement of the singular elastic crack tip in our native Abaqus workflow, classified as DOCUMENTED_UNRESOLVABLE_FROM_PUBLISHED_INFORMATION.
4. **Task 6 (Companion Visualization Bridge):** COMPANION_VISUALIZATION_BRIDGE_FULLY_VERIFIED under Job 1400408.mmmaster02 with 0.000000% mechanical parity. Formal authentic tool integration remains TASK6_BLOCKED_EXTERNAL_ABAQUSER_DEPENDENCY, meaning Task 6 is not complete.

Chapter 8

Conclusions, Project Status, and Next Work

8.1 Summary of established scientific results

The research conducted through 02 September 2026 has established the following verified findings:

1. **Fixed-Mesh Reference Reproduction (Task 3):** Reconstructed the Mode-I tensile fracture benchmark of Pandey & Kumar (2025) on a standard uniformly refined mesh (15,192 physical Quad4 elements, 15,305 nodes, Job 1398090.mm.master02, walltime 06 h 31 min, CPUT 06 h 30 min), achieving $F_{\text{peak}} = 0.757778 \text{ kN}$ at $u_{\text{peak}} = 0.005857 \text{ mm}$ (-0.029% force error, -0.051% displacement error vs published target). This validated the underlying user-element formulation and established the quantitative baseline.
2. **Abaqus-Native Adaptive Refinement Implementation (Task 4):** Fully automated the Python-driven two-job pre-refinement workflow using Abaqus recovered stress indicator MISESERRI, RemeshingRule, and adaptiveRemesh(odf=...). Resolved an implementation defect where calling generateMesh() had overwritten adaptively refined meshes, and verified deterministic reconstruction into coincident phase UEL, mechanical UEL, and facsimile UMAT layers.
3. **Forensic Reconciliation of the Nominal-1% Discrepancy (Task 5):** Proved mathematically and numerically that because the relative stress error $\eta = \text{MISESERRI}/\bar{\sigma}$ is scale-invariant across a singular elastic crack tip, applying the published nominal errorTarget = 1.0% marks $> 80\%$ of the domain for refinement in our native implementation, producing 71,320 elements. While this explains our observed over-refinement, the cause of why the published text reported 13,941 elements cannot be uniquely determined and remains classified as DOCUMENTED_UNRESOLVABLE_FROM_PUBLISHED_INFORMATION.
4. **Authoritative Quantitative Adaptive Reproduction (Task 5):** Reconstructed the adaptive model at errorTarget = 2.0%, yielding 15,396 physical elements (+10.4% match to published $\approx 13,941$). Production Job 1400395.mm.master02 completed 7,028 increments with zero cutbacks (Exit status 0, walltime 07 h 06 min, CPUT 06 h 52 min), achieving $K_0 = 137.8437 \text{ kN/mm}$ (-0.08%), $F_{\text{peak}} = 0.748197 \text{ kN}$ (-1.29%), $u_{\text{peak}} = 0.005775 \text{ mm}$ (-1.45%), and a 98.31% post-peak load drop,

successfully passing all five scientific acceptance gates (QUANTITATIVE ADAPTIVE REPRODUCTION PASSED / SCIENTIFICALLY ACCEPTED).

5. **Controlled 5.0% Sensitivity Evaluation (Task 5 / Task 7):** Reconstructed the adaptive model at `errorTarget = 5.0%` (4,194 physical elements, Job 1400396.mmmaster02, walltime 01h55min, CPUT 01h50min), demonstrating that coarse crack-corridor refinement ($h_{\min} \approx 0.00136$ mm) diffuses the phase-field damage gradient, shifting peak displacement to $u_{\text{peak}} = 0.007060$ mm (+20.48%) and classifying this run as **SCIENTIFICALLY_EVALUATED**.
6. **In-Solver Companion Visualization Bridge (Task 6):** Executed production Job 1400408.mmmaster02 (walltime 07h23min, CPUT 07h08min) coupling UEL phase and history fields to passive facsimile UMAT elements. Proved exact 0.000000% mechanical parity across all 7,028 increments (zero parasitic stiffness). Identified bounded phase overshoot ($\max(d) = 1.00518$) as **FORMULATION_LEVEL_1** arising from the continuous unconstrained phase equation (**POSSIBLE_FORMULATION_DISCRETI**).
7. **External ABAQUSER Dependency Boundary (Task 6):** Confirmed that the authentic IMFD ABAQUSER post-processing utility requires internal institute access and is not present in the current environment (**TASK6_BLOCKED_EXTERNAL_ABAQUSER_DEPENDEN**). Because authentic integration is pending, Task 6 is not complete. Prepared formal re-entry checklists and supervisor request packages.
8. **State Transfer and HPC Governance:** Demonstrated nonmatching state mapping and runtime UEXTERNALDB ingestion, identified mechanical relaxation challenges across topology splits, and established the single-core serial execution boundary for mutable common blocks.

8.2 Authoritative thesis task status

Table 8.1 defines the authoritative status of all thesis proposal work packages as of 02 September 2026.

Table 8.1: Authoritative status matrix of Master’s thesis work packages.

Work age	Pack-	Scope / Description	Governed Status	Evidence / Refer- ence
Task 1		PFF familiarization	COMPLETE	Documented
Task 2		Literature review	COMPLETE	Ongoing writing sup- port
Task 3		Reference reproduction w/o refine- ment	COMPLETE / AC-	Job
Task 4		Implement native refinement in Python	CEPTED COMPLETE / VERI-	1398090.mmmaster02
Task 5		Reference reproduction with refine- ment	FIED QUANTITATIVE	Job
Task 6		Integrate ABAQUSER visualization	PASSED COMPANION	1400395.mmmaster02
Task 7		Fracture benchmarks & sensitivities	VERIFIED / EXT.	Job
Task 8		Meshing-parameter recommendations	BLOCKED	1400408.mmmaster02
Task 9		Document workflow for future users	NOT STARTED	Ready to commence
Task 10		Write thesis manuscript	NOT STARTED	Follows Task 7
			NOT STARTED	Follows Task 8
			IN PROGRESS	Active manuscript

8.3 Roadmap for immediate next work

With Task 5 quantitatively validated and Task 6 clearly established at the companion-bridge level, the workflow proceeds along the following parallel paths:

1. Supervisor Action Item for Task 6:

- Obtain the authentic ABAQUSER post-processing utility from IMFD / supervisor;
- Upon receipt, execute the tool directly in post-processing on the verified ODB artifact (PK_MODE1_PROPOSED_PFM_VIS.odb) to close the external dependency;
- Benchmark the reconstructed ODB output against the verified companion bridge contours.

2. Immediate Technical Advancement to Task 7:

- Conduct controlled mesh-refinement sensitivity studies across error targets (2.0%, 5.0%, and intermediate values) to establish accuracy-versus-cost Pareto frontiers;
- Perform load-increment sensitivity studies holding the physical mesh fixed while evaluating convergence thresholds;
- Apply the verified adaptive remeshing methodology to secondary fracture benchmarks (e.g., Mode-II shear and asymmetric notched plates).

3. Advancement to Task 8:

- Synthesize sensitivity data to formulate quantitative meshing guidelines ($h_{\min}/\ell_0$, errorTarget, refinementFactor) for future engineering users.

Appendix A

Computational Provenance and Authoritative Execution Ledger

A.1 Authoritative HPC execution ledger

Table A.1 provides the complete, authoritative ledger of primary high-performance computing calculations executed under this research program, cataloged with their exact PBS identifiers, physical mesh discretizations, verified walltimes, and formal scientific classifications.

A.2 Cryptographic checksums of primary production packages

To ensure strict scientific reproducibility, SHA-256 cryptographic hashes for the primary production packages are recorded below:

A.2.1 Accepted Task-5 adaptive production solve (Job 1400395.mmaste02)

- Production Input Deck (PK_MODE1_PROPOSED_PFM.inp):
c745a42f2c60a82ae5b8b2bff38887a38f6e3b1570a7b2673af4178cbf677b0c
- User Subroutine (f42_mixed_uel.for):
5abf77b570c67283f082e02d8ba2fd0111db843ad9c4cd3501eb9aaaf149dfdd
- PBS Launcher Script (submit_solver.pbs):
7cb52ff6985a6a58bc2e22642142cbbef311c19b48cbe005b8da5aeb785a9bc

A.2.2 Verified Task-6 companion visualization solve (Job 1400408.mmaste02)

- Production Visualization Deck (PK_MODE1_PROPOSED_PFM_VIS.inp):
e29694cf60f47b2dceb84d904999715c5857e36eecdcccb5dbcdf17c89dbd07b
- User Subroutine (f42_mixed_uel.for):
9e6b5c97636c1c1b735a27507f372fbf36b70b8a8d49ad7903b3d237c8fee181
- PBS Launcher Script (submit_solver.pbs):
a4c51486d38e21a812dfceb65f3ad36398fe265fa1a07409f875bbef315df612

Table A.1: Authoritative execution ledger of primary HPC calculations through 02 September 2026.

PBS Job ID	Model / Case	Elements / Nodes	Walltime	Scientific Role	Classification &
1398090.mmaste02	Fixed-Mesh Ref.	15,192 / 15,305	06 h 31 m	SCIENTIFICALLY_ACCEPTED (Task 3: $F_{\text{peak}} = 0.7578$ kN, error -0.029%)	
1379893.mmaste02	Coarse Pre-Analysis	2,906 / 2,988	00 h 04 m	QUALIFIED (MISESRI extrac- tion, 3,930 rows)	
1398807.mmaste02	Blunt Notch 1%	74,261 / 73,806	12 h 15 m	TECHNICAL_PASS_SCIENTIFIC_FAIL (Rejected: incorrect notch geome- try)	
1398865.mmaste02	Sharp Seam 1%	71,320 / 71,490	10 h 45 m	TECHNICAL_TERMINATION (Aborted at $\Delta t_{\text{min}} = 10^{-9}$ limit)	
1399632.mmaste02	Sharp Seam 1%	71,320 / 71,490	35 h 08 m	QUANTITATIVE_REPRODUCTION_FAIL (Over-refinement compliance shift)	
1400367.mmaste02	Adaptive 2% DC	15,396 / 15,414	00 h 01 m	DATACHECK_PASSED_CLEAN (Exit status 0)	
1400368.mmaste02	Adaptive 2% Solve	15,396 / 15,414	22 h 10 m	UNCORRECTED_SUPERSEDED (Viscous rate-dependent hardening artifact)	
1400381.mmaste02	Adaptive 5% DC	4,194 / 4,274	00 h 01 m	DATACHECK_PASSED_CLEAN (Exit status 0)	
1400382.mmaste02	Adaptive 5% Solve	4,194 / 4,274	05 h 42 m	UNCORRECTED_SUPERSEDED (Viscous rate-dependent hardening artifact)	
1400395.mmaste02	Adaptive 2% Final	15,396 / 15,414	07 h 06 m	SCIENTIFICALLY_ACCEPTED (Task 5: $F_{\text{peak}} = 0.7482$ kN, error -1.29%)	
1400396.mmaste02	Adaptive 5% Final	4,194 / 4,274	01 h 55 m	SCIENTIFICALLY_EVALUATED (Sensitivity point: $F_{\text{peak}} =$ 0.7650 kN, u_{peak} delay $+20.48\%$)	
1400408.mmaste02	Companion 2% Vis	15,396 / 15,414	07 h 24 m	COMPANION_BRIDGE_VERIFIED (Task 6: 0.000000% RF parity vs 1400395)	
1379393.mmaste02	Mode-II H0	3,930 / 4,082	00 h 17 m	BENCHMARK_EVALUATED (Mode-II shear baseline)	
1379433.mmaste02	Mode-II H1	12,064 / 12,382	00 h 43 m	BENCHMARK_EVALUATED (Mode-II intermediate mesh)	
1379966.mmaste02	Mode-II H2	33,852 / 34,390	04 h 43 m	WALLTIME_CENSORED (Killed at standard queue walltime limit)	

A.2.3 Authoritative Task-3 fixed-mesh reference (Job 1398090.mmaster02)

- Standard Reference Input Deck (PK_MODE1_STANDARD_PFM.inp):
ad7310574f26b527815cfc876b6b7115858fc85775f0f3532cf2b60ca7bb59bf
- User Subroutine (f42_mixed_uel.for):
5abf77b570c67283f082e02d8ba2fd0111db843ad9c4cd3501eb9aaaf149dfdd

A.3 HPC notification status record

- Telegram Webhook Trap: Verified operational across all production jobs (1400368, 1400382, 1400395, 1400396, 1400408).
- PBS Email Delivery: Direct cluster SMTP email notification failed to deliver due to local relay limitations and remains cataloged as an infrastructure defect.

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